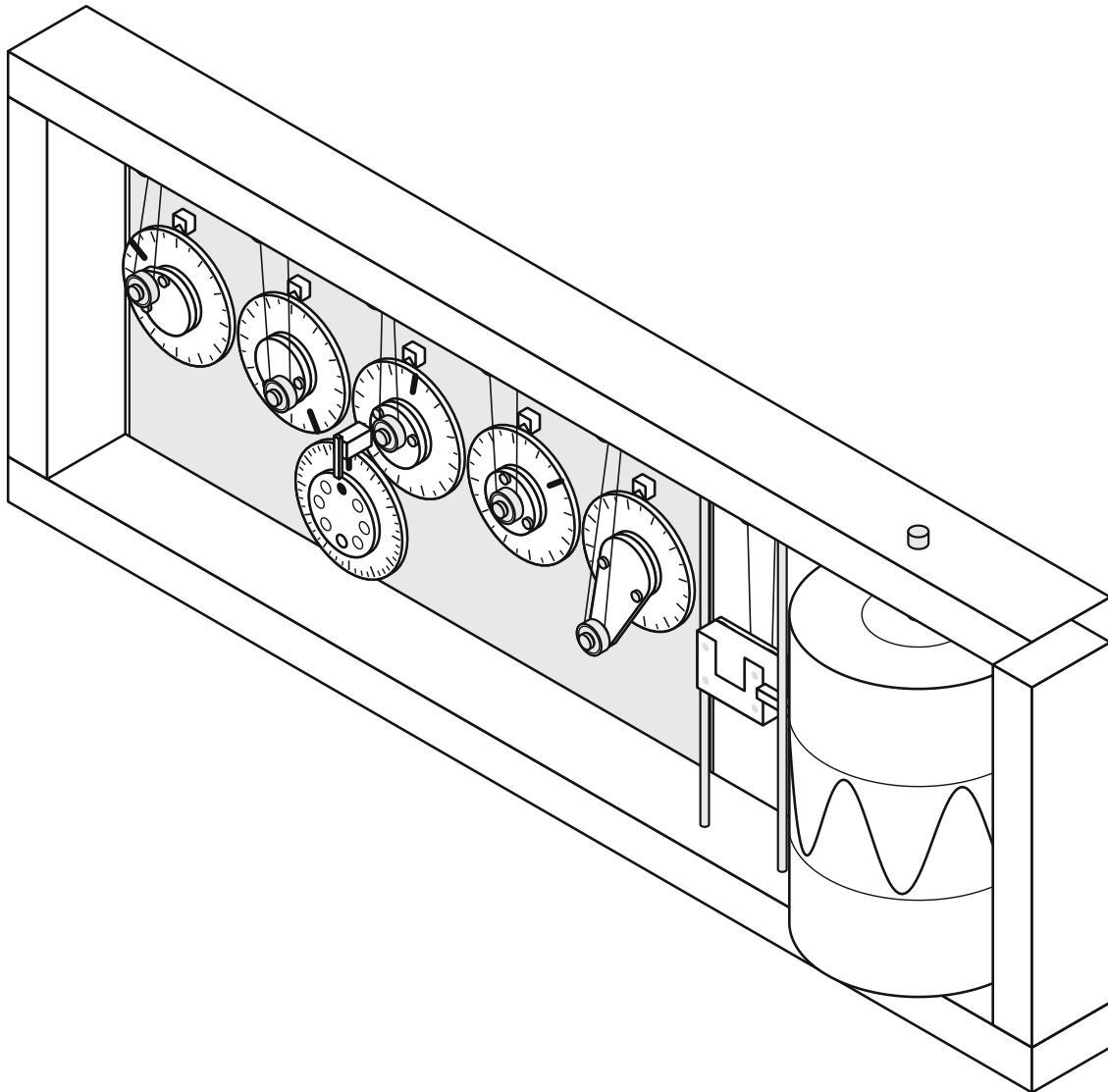


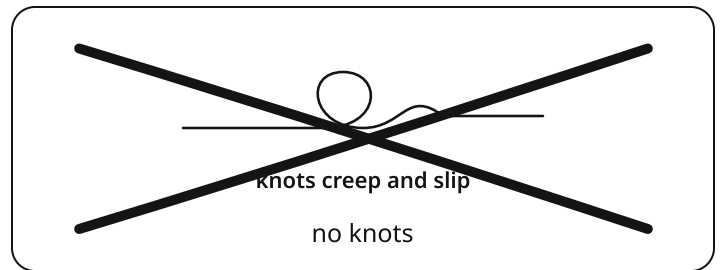
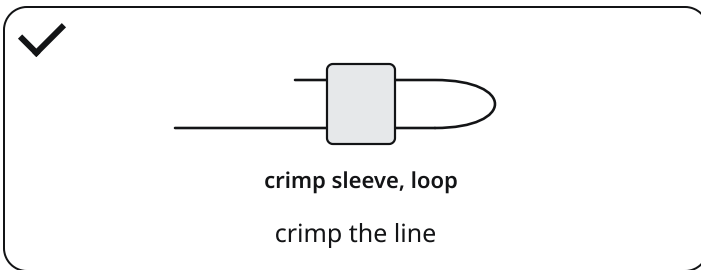
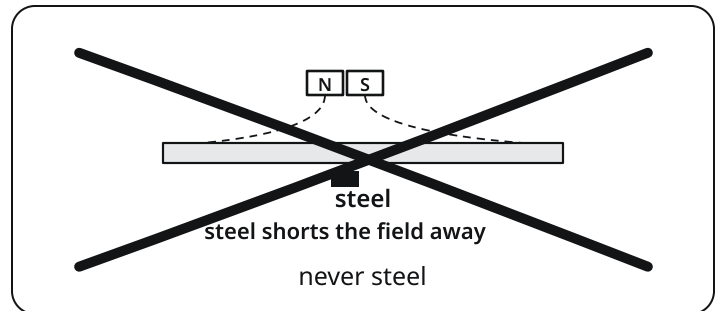
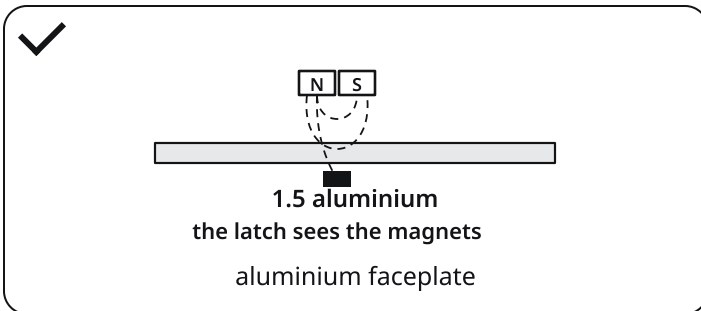
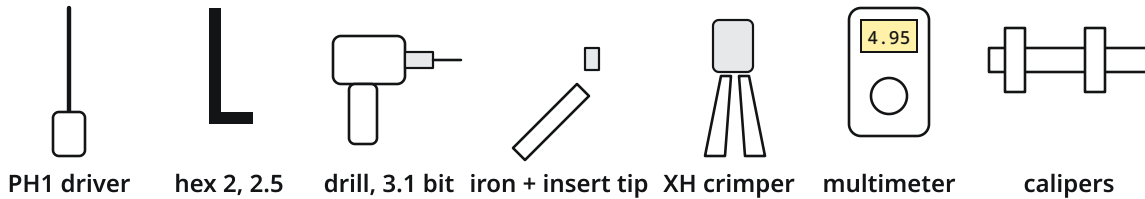
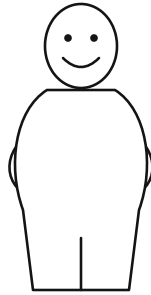
Tide Machine

A five-constituent harmonic tide predictor: five stepper-driven arms sum on one line and draw the tide on a drum.

Assembly, board, smoke test, wiring, first power.



Design and Quality
the bench of Mike



i **Datum.** Every number is in the machine frame: **x** along the row (S2 at -224 ... M2 at 0, pen +70, drum +155.5), **y** up, **z** negative toward you. The faceplate's front face is $z = 0$. The figures are drawn from `station_study.py`, the same model the fit checks run on.

! **Never turn an arm at constant speed**, not even to test it by hand with a drill: the firmware drives each arm at an uneven rate so the line length is exactly sinusoidal.

Contents

A Mechanics

steps 1 to 16

B Main board

steps 17 to 22

C Smoke test

steps 23 to 29

D Wiring

steps 30 to 36

E First power-up, calibration

steps 37 to 45

F How to use

steps 46 to 49

G Appendix: schematics

system, logical, full board

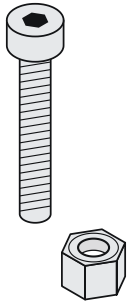
! Known issues

read before you order or cut

Give feedback by **step number**; they won't change. Status: the mechanics are Rev M (fit-checked, not yet built), the board is rev B (routed, DRC clean, not yet ordered), and the firmware isn't written, so section E is mostly **planned**.

1

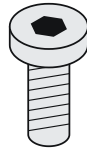
Fasteners



6x

pulley-bracket axles

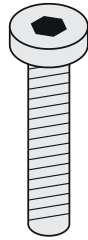
M3 x 20 + nyloc



5x

arm-bearing axles

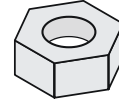
ISO 7380 M3 x 8



5x

pen pulley + 4 carriage axles

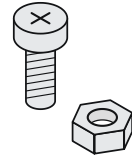
ISO 7380 M3 x 16



4x

carriage axles, rear face

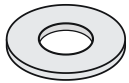
ISO 4032 M3



12x

motor ears; head 2.4 tall or less

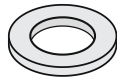
M3 x 8 pan + nut



12x

under the ear nuts only

M3 washer



16x

brackets 6, carriage 4, arm axles 5, pen pulley 1

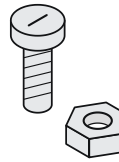
3 x 5 x 0.5 shim



6x

5 arms + carriage front

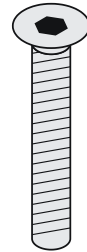
M3 x 4 heat-set



30x

arms, Hall and calendar carriers, tabs, hand

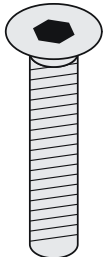
M2 x 6 + nut



12x

pulley brackets

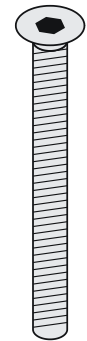
#4 x 3/4" flat



10x

faceplate to the rails

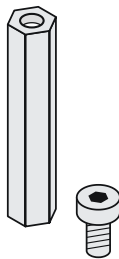
#6 x 3/4"



4x

end posts (KI 9)

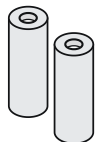
#6 x 1-1/2"



4x

main board standoffs, 8 screws

M3 x 30 F-F + M3 x 6



4x

line ends

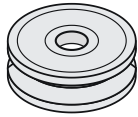
crimp 0.6-0.8 ID



Shims, not washers, against a bearing. An M3 washer is Ø7 and presses on the V623ZZ's shield; the inner race is only Ø5. Every bearing face that gets clamped gets a 3 x 5 x 0.5 shim. That's 16, one more than the 15 the BOM lists (KI 6).

2

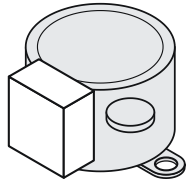
Motion, sensing, frame



22x

12 rail, 5 arms, pen pulley, 4 carriage

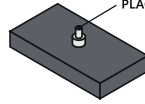
V623ZZ 3 × 12 × 4



6x

five stations + the drum

288YJ-48 5 V



1x

calendar, dual concentric shafts (**KI 5**)

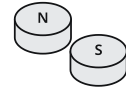
PLACEHOLDER
BKA30D-R5



7x

Hall latches: 5 stations, year, moon

13AG
1 2 3
DRV5013 TO-92



14x

seven opposed pairs

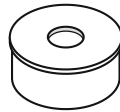
N52 5 × 2 axial



2x

pen guide rods

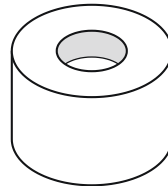
3 mm h6 × 170



2 m

the line; not nylon

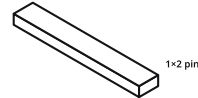
0.30 fluorocarbon



130

the drum, cut square; 7 mm shaft

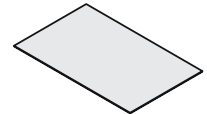
3" Sch 40 PVC



1302

rails 2 × 499, posts 2 × 152

1 × 2 pine
1+2 pine



1x

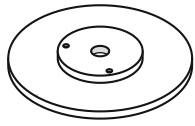
faceplate 283 × 176, laser cut (**KI 1**)

5052 Al 1.5

Also: perfboard (70 × 90, double-sided) for the carriers, 3-core 26 AWG and 8-core lead, JST XHP housings and crimps, PLA, clear silicone (fallback only), decal vinyl, a 0.3-0.5 mm fibre-tip pen.

3

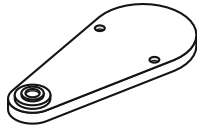
Printed parts, as they sit on the bed



hub_dial.stl

5x

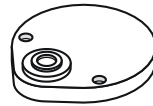
dial back down, boss up



arm_M2.stl

1x

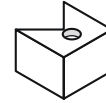
r 35.00; back down



arm_S2 / N2 / K1 / 01.stl

4x

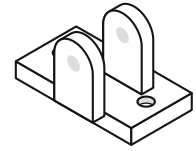
r 8.75, 7.35, 3.50, 2.45



index_tab.stl

5x

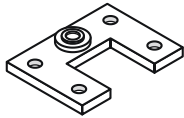
flat



pulley_bracket.stl

6x

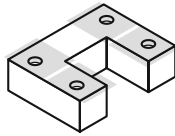
rail face down



pen_carriage_rear.stl

1x

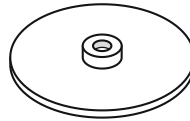
pulley stub up



pen_carriage_front.stl

1x

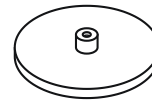
pockets open upward



year_dial.stl

1x

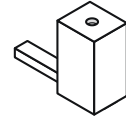
decals face down (KI 5)



moon_disc.stl

1x

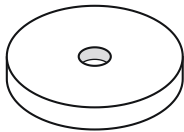
decals face down (KI 5)



calendar_hand.stl

1x

front face down

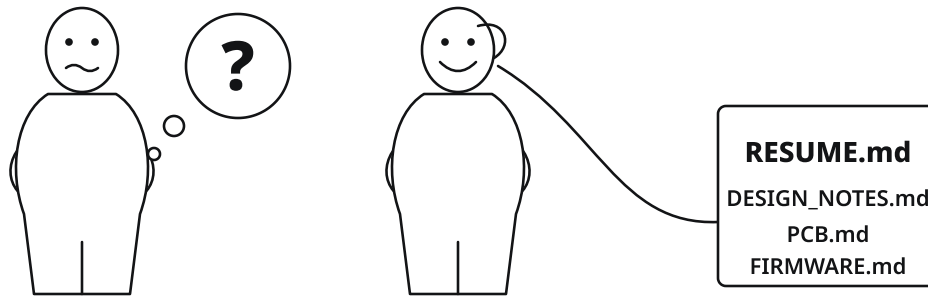


drum_end_cap.stl

2x

flat (KI 8)

 **Print the coupons first:** the D-bore (tune it to a light press on a motor's flats), a magnet pocket at 5.1 and 5.2, the 4.0 insert hole, and a PVC-bore ring for the drum cap. PLA, 3 perimeters, no supports anywhere. The files are in `stl/`, regenerated by `export_stl.py` from the same model as these drawings.



Known issues found while writing this guide

check before you order or cut

- KI 1 No Rev M cut files yet.** `tide_machine.rb`, `faceplate.py` and `rails.py` still describe Rev L. The hole schedule in step 1 comes from `station_study.py`. Port it to the Ruby before ordering the plate or drilling the rails. The #6 frame-screw holes ($\varnothing 4.0$ at $y \pm 82$) have no spacing yet. Step 1, step 14.
- KI 2 Hall windows are still sized for the SOT-23.** They're 3.4×3.4 ; the TO-92 latch needs about 4.4×3.4 , and its sensing element sits a few tenths deeper. Re-run the field check (the magnet gap was 4.40 mm) when you resize them. Step 1, step 5.
- KI 3 No holes for the main board yet.** Proposed: $4 \times \varnothing 3.2$ at (25, 39), (-108, 39), (25, -35.28), (-108, -35.28), for the board upright as seen from behind. Checked against the whole model (the standoffs, the board and the lead ends under it): nothing within 3 mm. Screw heads under a dial are fine at 2.4 mm tall or less (the same 1.1 mm as the motor-ear screws). Step 30.
- KI 4 The Hall carriers and the calendar carrier are hand-cut perfboard**, not designed parts. As a plain 36×68 rectangle, the calendar carrier's top corners clipped the N2 and K1 motor bodies (8.9 mm^3 each; each corner is 11.7 mm from a motor's axis, against its 14 mm radius). **Fixed in the model:** cut 4×6 off both top corners; it now clears by 0.98 mm. **Still open:** the two bottom holes at $y -78$ are under the bottom rail (its inner face is at $y -76$), so their M2 heads have nowhere to go on the front; and all four nuts sit inside the BKA placeholder's outline. Move the holes once the motor is measured. Step 5, step 6.
- KI 5 BKA30D-R5 is a placeholder.** The body, the shaft diameters, lengths and positions, and whether both shafts turn continuously (the family datasheet shows stops) are all unconfirmed. The year dial's and moon disc's hub bores are placeholders, and so is the coil-to-pin map in step 34. Step 6, step 10, step 34.
- KI 6 Arm-bearing axles clamped a washer on the shield.** The $M3 \times 8$ went through an M3 washer ($\varnothing 7$) against the V623ZZ's face; the inner race is $\varnothing 5$, so 1 mm of the washer bore on the shield. The pen pulley's $M3 \times 16$ head ($\varnothing 5.7$) did the same by 0.35 mm. **Fixed in the model:** a $3 \times 5 \times 0.5$ shim under each (16 shims in all). The pen pulley's screw now engages its insert by 2.25 mm. Step 9, step 12.
- KI 7 The pen arm ran through the +x guide rod** (27.9 mm^3) at the drum's mid-plane ($z -22$ to -26 , rod at -23.75). **Fixed in the model:** the arm runs 1 mm in front of the rods, at $z -26.25$ to -30.25 ; the nib meets the drum at $x 111.2$. The pen holder, leaf spring, retract lever and ballast pocket still aren't designed. Step 12.
- KI 8 Drum drive not designed.** Nothing is drawn yet for the motor mount, the coupling to the 7 mm shaft, or how the end caps grip the shaft (the cap STL only has a $\varnothing 7.1$ bore). The drum motor sits

outside the top rail, about 18 cm in a straight line from J10; its lead is about 24 cm. Route it tight or make a 5-way XH extension. Step 15, step 32.

KI 9 End-post screws are too short. The BOM's #6 × 3/4" can't reach through a 19 mm rail into the post's end grain. Use #6 × 1-1/2" (4) with glue; the faceplate still takes #6 × 3/4" (10). Step 13.

KI 10 The far dead end has no fixing. Nothing slides at S2's IN pulley, so the line can end there, but no tie-off point is designed. The pen bracket's screw-adjustable anchor, copied onto S2's bracket, would do. Step 16.

KI 11 Decals use placeholder constituents. Put NOAA G values, the longitude and the epoch into `decals.py`, and check the K1/O1 ±90° convention (it moves those zeros by 12 h), before printing. Step 8.

KI 12 No firmware yet. Everything in section E and F tagged **planned** needs it: the inverse tables, the Hall edge timing, homing, NTP, the PWM hold. Section E.

KI 13 No 1:1 rail templates yet. `rails.py` printed drilling templates for Rev L; Rev M needs new ones from the model, printed at 100% and tiled across Letter pages (the rail is 498.9 long), with a calibration bar, page-join marks, and each sheet labelled top or bottom rail, face, and S2 end. Until then, measure from the **S2 end**. Top rail, inner face: #4 pilots at 50, 106, 162, 218, 274 and 344 (each ± 11), 15 from the rear edge; rod sockets Ø3.1 × 9 at 325.08 and 362.92, 23.75 from the rear edge (the bottom rail's inner face takes the same two). Both rails: the drum hole at 429.45, 24 from the rear edge. Rear faces: faceplate pilots 6 from the inner edge (spacing with KI 1). Outer faces: end-post screws 9.5 from each end. Step 11, step 13.

Fixed in the model while writing this guide

- The calendar sat at x -196 in the fit study but -140 in the spec; now -140 everywhere.
- The left end post ran into S2's dial, magnets and ear screws; the rails grew 12 mm (now 499; left post 5 mm clear of the dial).
- The calendar index hand's base overlapped N2's dial; it was narrowed and moved down (1.09 mm clear).
- The index tab and the calendar hand had no pilot holes for their M2 screws; both STLs now have a Ø1.6 pilot.
- KI 4, KI 6 and KI 7 above. Each fix was confirmed with a real interference check (manifold intersections of every pair of parts, `check_interference.py`) and the fit study's 31 clearances, all ≥ 0.98 mm.

A Mechanics

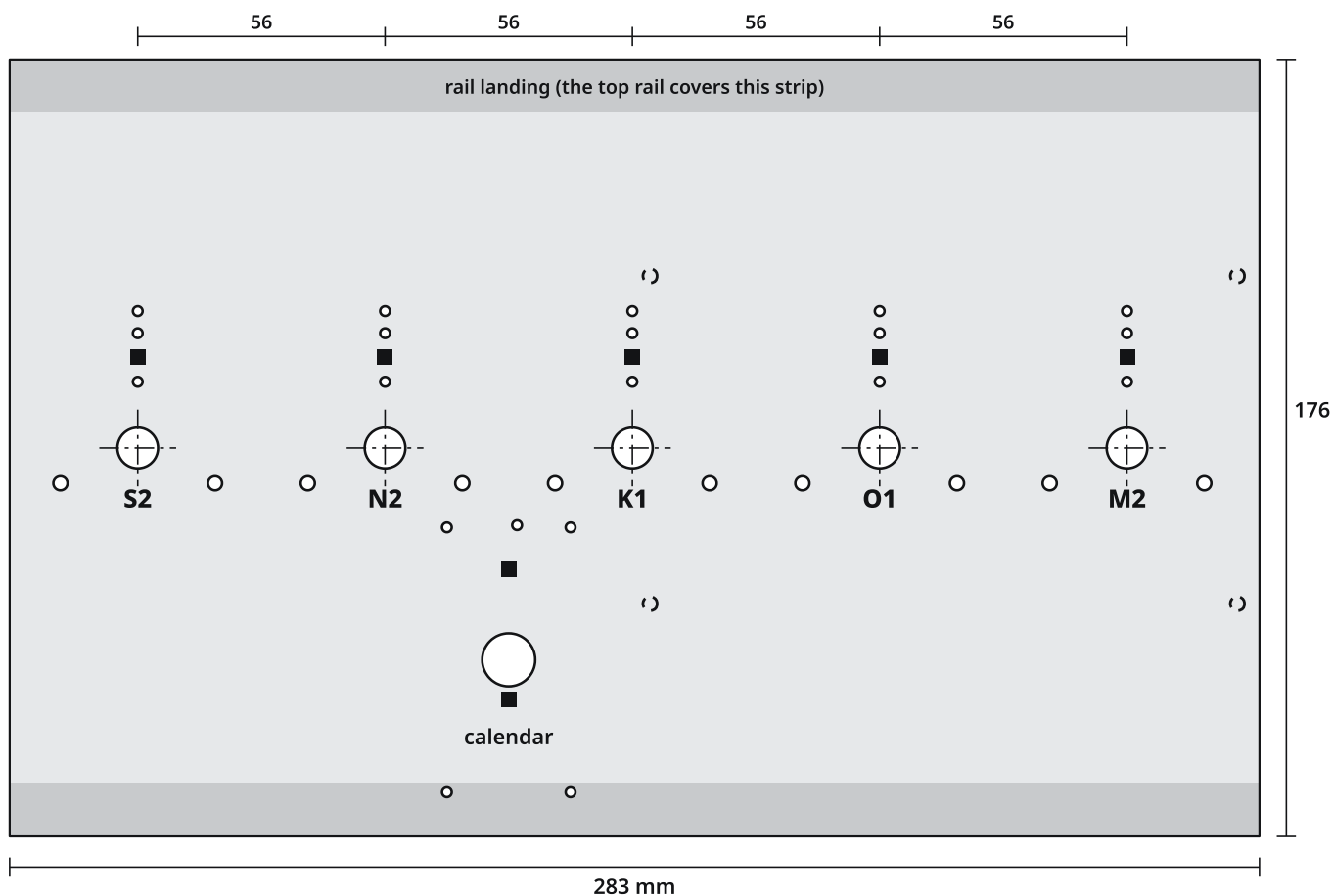
Plate work first: behind it (motors, carriers), then in front (dials, arms). Then the frame, with the pen carriage inside it, and the plate onto the frame. The line goes on last.

1 Check the faceplate

KI 1

KI 2

KI 3



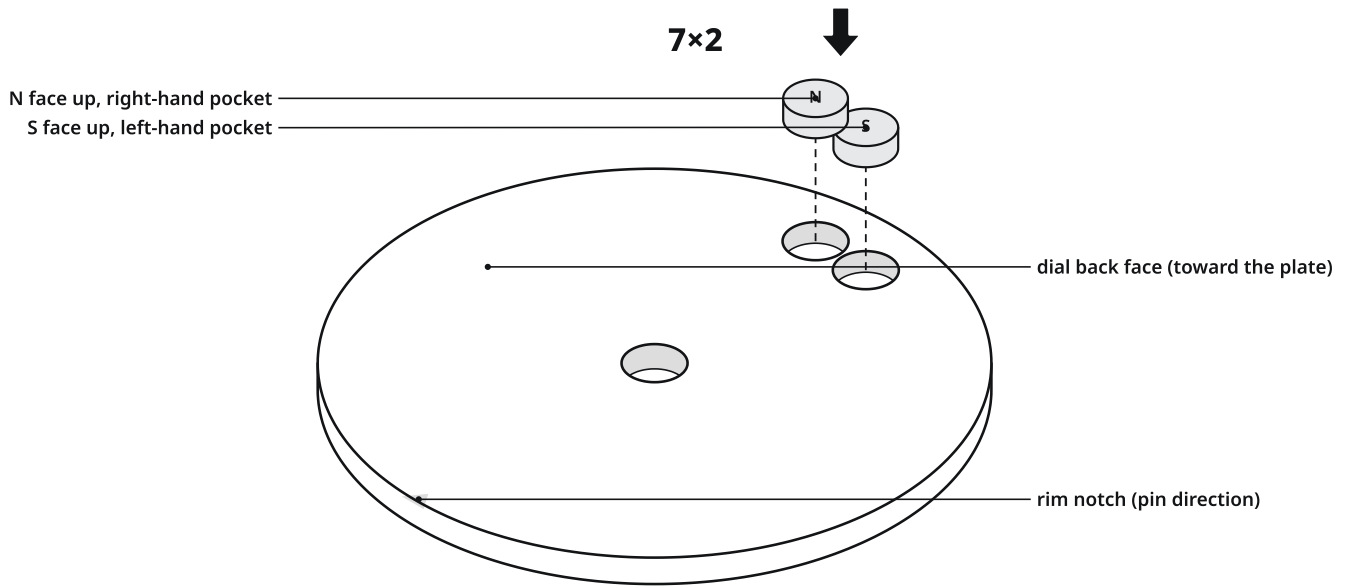
seen from the FRONT. dashed circles: proposed main-board standoff holes (KI 3)

Count	Size	Where (per station at shaft x, 0)	For
5	Ø9.2	(x, 0)	locates each motor's Ø9 boss
10	Ø3.2	(x ± 17.5, -8)	motor ears, M3
5	window 3.4 (KI 2)	centred (x, 20.5)	the Hall latch nests in it
10	Ø2.2	(x, 15) and (x, 26)	Hall carrier, M2
5	Ø2.2	(x, 31)	index tab, M2
1	Ø12	(-140, -48)	calendar shafts and hub adapters
2	window 3.4	(-140, -27.5), (-140, -57)	year and moon latches
4	Ø2.2	(-140 ± 14, -48 ± 30)	calendar carrier
1	Ø2.2	(-138.1, -17.5)	calendar index hand
4	Ø3.2, proposed	(25, 39), (-108, 39), (25, -35.28), (-108, -35.28)	main board standoffs (KI 3)

- Measure the **1.5 mm aluminium** plate against this schedule before anything else goes on it: the pitch is **56.0** between every pair of shafts.
- With the row reading S2 → M2 left to right from the front, the calendar cluster sits **below, between N2 and K1**. A mirrored plate is scrap.

 **Never steel.** The Hall latches read the magnets through the plate; steel would short the field away.

2 Glue the magnets into the dials

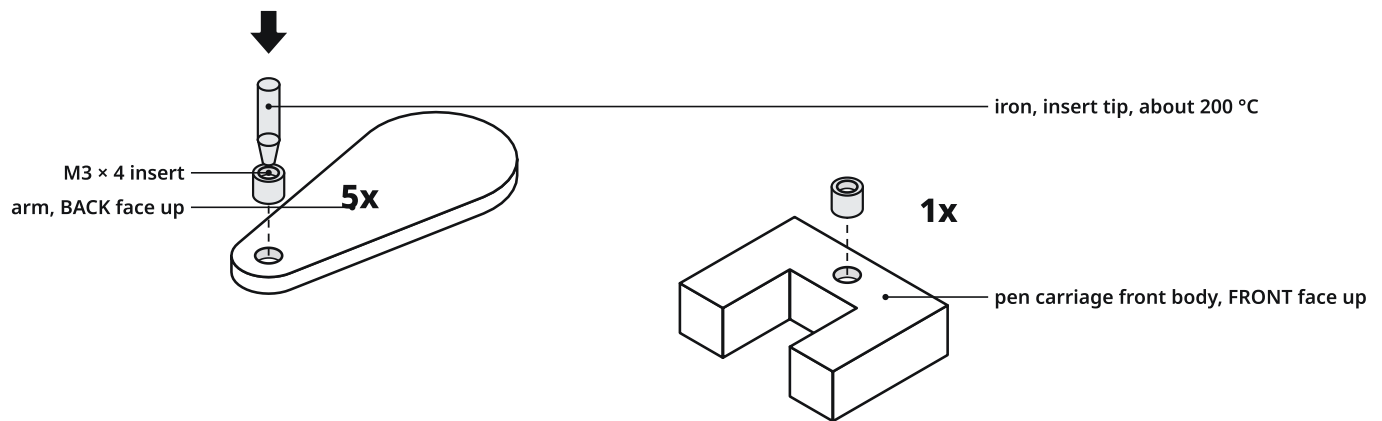


- Turn each `hub_dial.stl` over, back face up. The two 5×2 pockets sit at $r\ 20.5$, opposite the rim notch.
- Seen from the back with the notch at 12 o'clock: **N face up in the right-hand pocket, S face up in the left.** A drop of CA in each pocket.
- Same on the year dial (pockets at $r\ 20.5$) and the moon disc ($r\ 9$): seven pairs in all.

i The latch switches where the field flips sign between the two magnets, so the edge hardly moves with gap or temperature (0.15° per mT of drift). The same order on every dial makes every station report the same edge.

💡 Check each face with a spare magnet before gluing: one face pulls it, the other pushes it away.

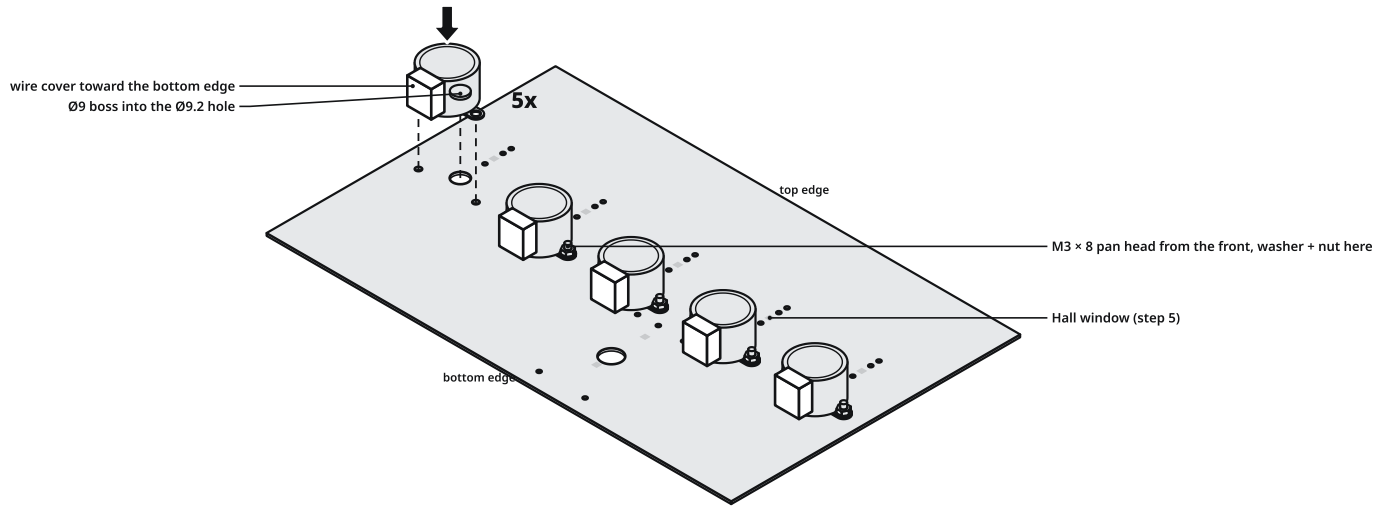
3 Press in the heat-set inserts



- Each of the five arms: **from the back face**, into the Ø4.0 hole under the bearing boss, flush.
- The pen carriage front body: **from the front face**, into the Ø4.0 hole at its centre. That's the pen pulley's screw thread (step 12).

 Keep the insert square and stop at flush. An insert standing proud on an arm's back face tilts the arm against the dial boss.

4 Bolt the motors to the back of the plate



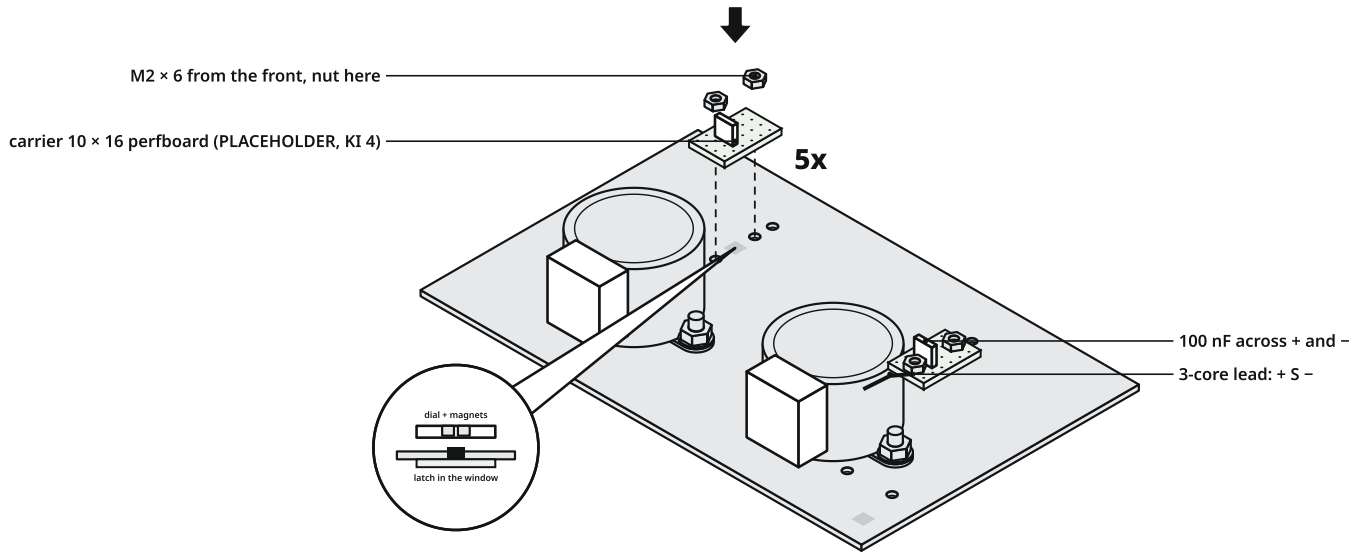
- Plate face down. Each 28BYJ-48: **boss into the Ø9.2 hole**, wire cover toward the plate's **bottom edge**.
- **M3 × 8 pan head** from the front through each ear, then a washer and a nut behind. Snug only.

i The plate locates each shaft by its boss; the ear bolts only clamp. An M3 in the Ø4.2 ear hole would let the motor wander ±0.6 mm. The pan heads stand 2.4 proud of the front face and the dials clear them by 1.1: longer heads won't fit.

5 Fit the Hall carriers

KI 2

KI 4



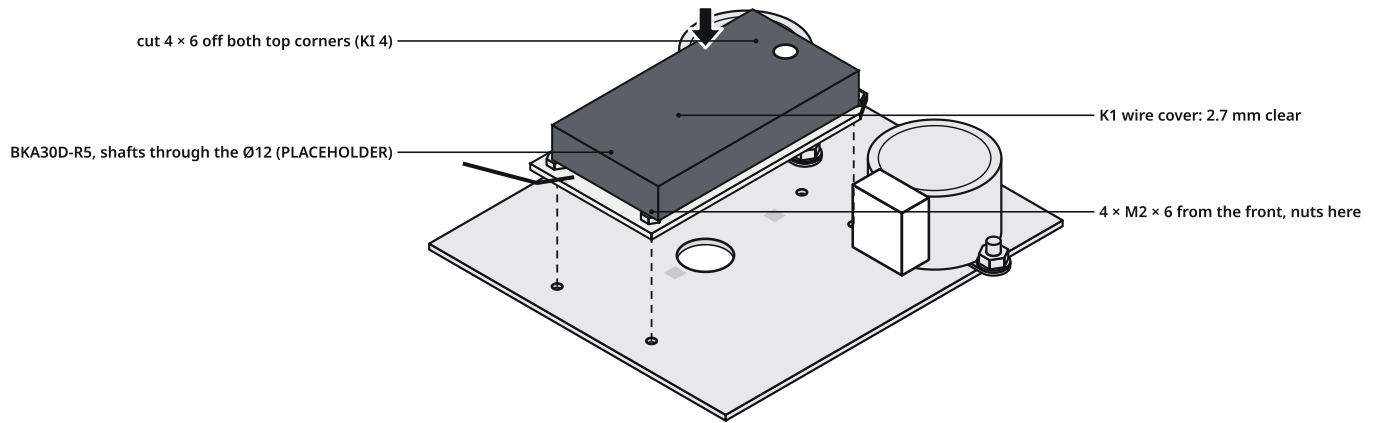
- Cut five carriers, 10 × 16, from perfboard. On each: the DRV5013 lying flat on the front face, branded face toward the plate, a 100 nF across + and -, and a 3-core lead. Label each lead with its station.
- The latch nests in its window at (x, 20.5). Two M2 × 6 from the front through (x, 15) and (x, 26), nuts behind.

 The DRV5013 in TO-92 has pin 1 VCC, 2 GND, 3 OUT: the carrier crosses 2 and 3 onto the lead's + S - order. Check against the datasheet for your variant.

6 Fit the calendar carrier

KI 4

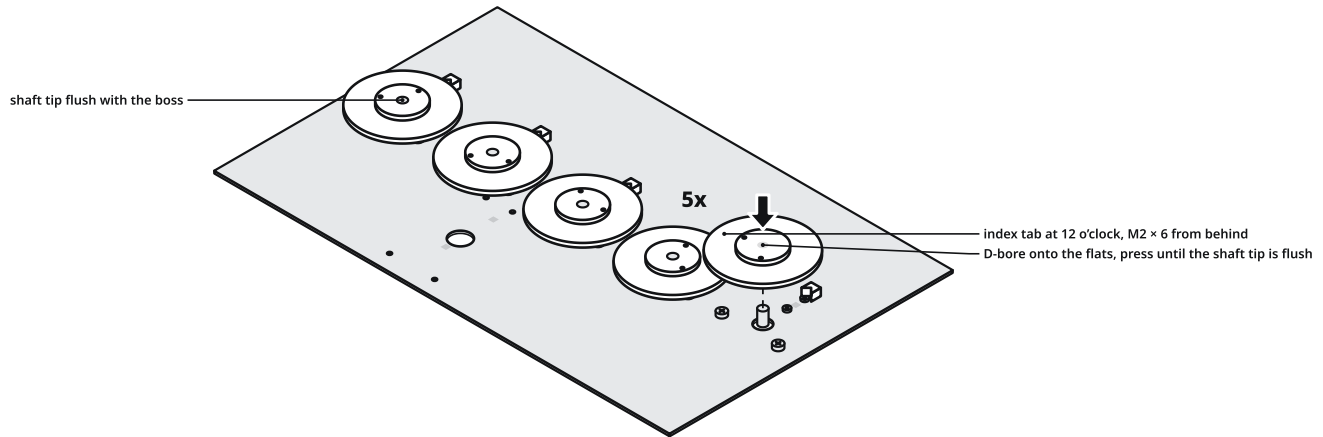
KI 5



- Cut the carrier 36 × 68 from perfboard, then **cut 4 × 6 off both top corners**. Without the cut it hits the N2 and K1 motor bodies.
- On it: the BKA30D-R5, shafts forward through the plate's Ø12, and the year and moon latches at (0, +20.5) and (0, -9) from its centre, nested in their windows.
- Four **M2 × 6** from the front, nuts behind. Then an 8-core lead for the coils (step 34), and a 3-core lead from each latch.

7

Press the dials onto the shafts; fit the index tabs

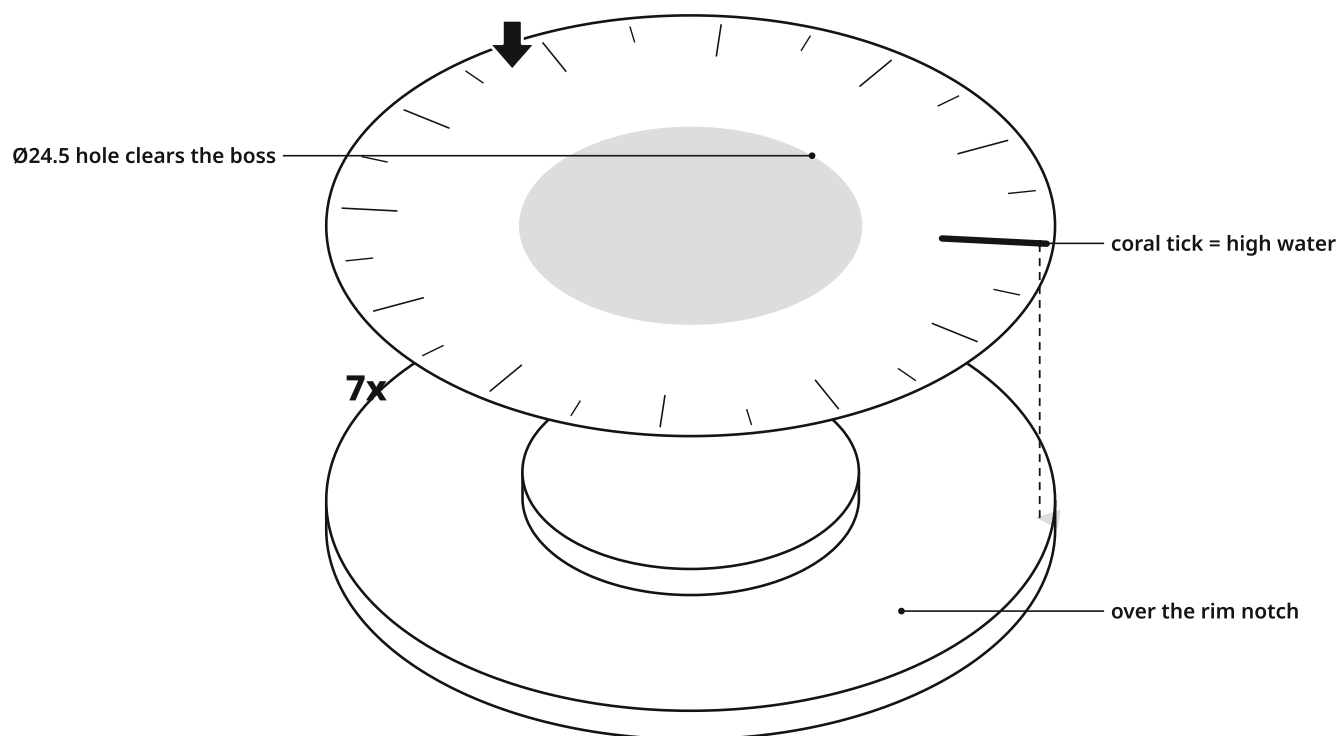


- Plate face up. Each hub + dial: the D-bore over the shaft's flats, and press **until the shaft tip is flush with the boss**. The dial's back is then 3.5 mm off the plate.
- An index tab at 12 o'clock above each dial, one **M2 × 6** from behind into its pilot.
- Turn each dial by hand: it must clear the ear-screw heads, the tab and its neighbours (4.0 mm apart).

i The D-bore carries the torque and a light press holds it: the line pulls in the dial's plane, so nothing pushes it off. For a loose one, a dab of clear silicone. Not threadlocker, which barely cures against plastic.

8 Apply the decals

KI 11



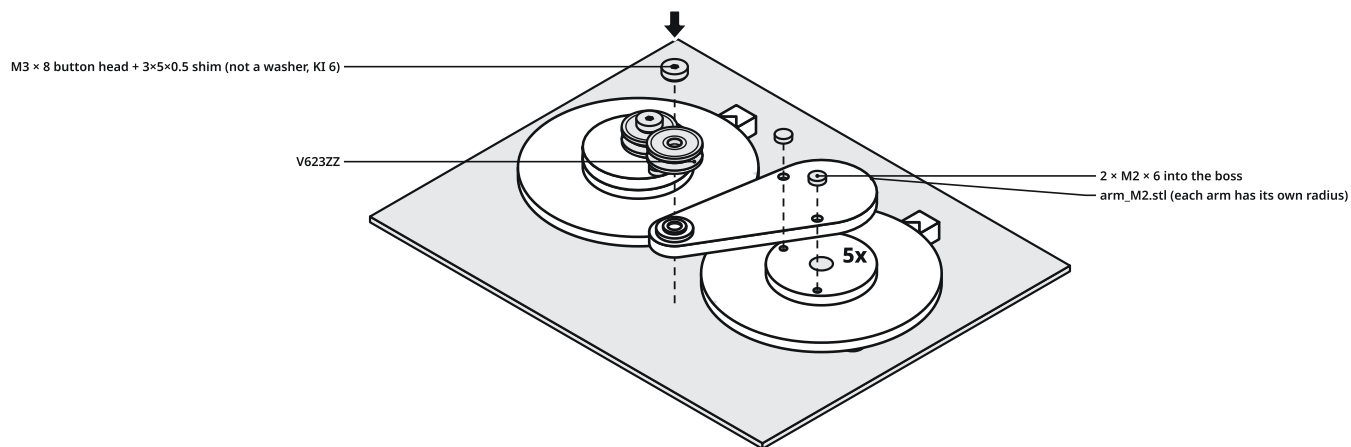
- Print `decals_1to1.pdf` at 100% on laser vinyl. Check the scale bar first.
- Each station decal (Ø52, Ø24.5 hole): **coral tick over the rim notch**. Year decal on the year dial, moon decal on the moon disc.

i The scales are warped: each tick sits where the arm really is (M2's ticks move up to ± 33 min). The decals depend on the constituents, the longitude and the year, so they're reprinted for each deployment.

9

Bolt on the arms and their bearings

KI 6



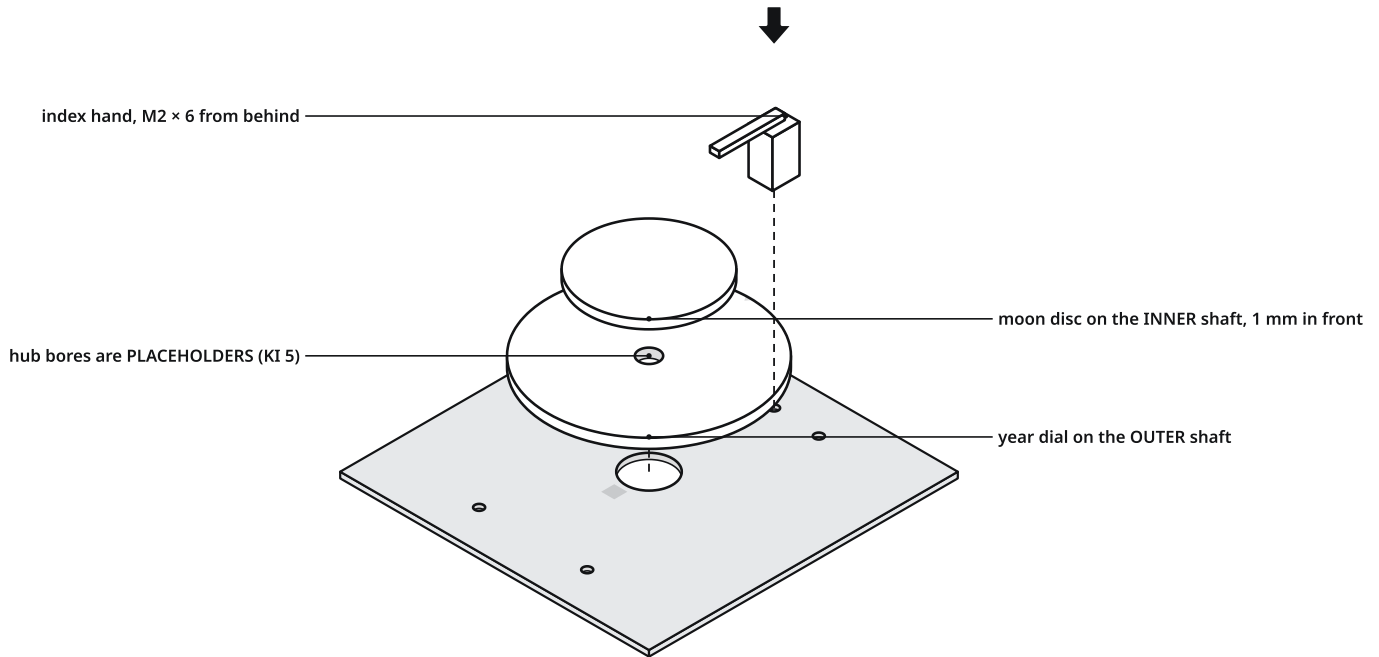
- Each arm onto its own dial: `arm_S2` ... `arm_M2`, stamped by radius. Two `M2 x 6` into the boss's pilots.
- A `V623ZZ` onto the stepped boss, a **3 x 5 x 0.5 shim**, then an `M3 x 8 button head` into the insert. Firm: the inner race is clamped, the outer spins.
- Spin each bearing: free, no scrape. M2's arm passes 2.5 mm over O1's dial.

i When real constituents replace the placeholders, only the arms are reprinted: the pin radius is the amplitude (35 mm per metre of tide).

10

Fit the year dial, the moon disc and the index hand

KI 5

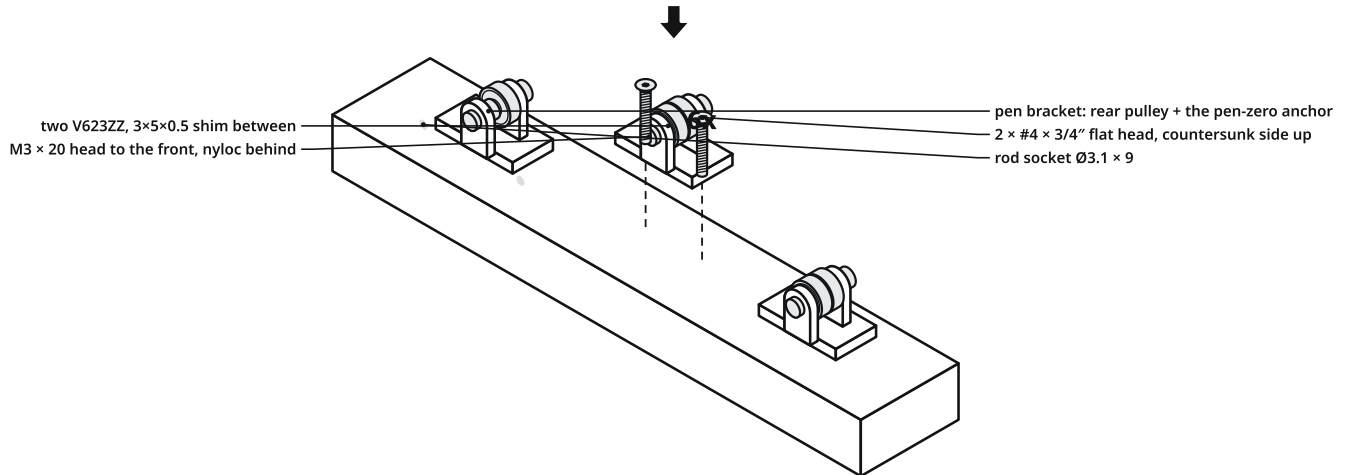


- Year dial onto the BKA30D-R5's **outer** shaft, magnets to the plate. Moon disc onto the **inner** shaft, 1 mm in front of the year dial.
- The index hand, one **M2 × 6** from behind: it crosses the year ring and stops at the moon disc's rim.
- Turn both by hand: the gauge motor is weak (a 2.5 g pointer rating), so the discs must spin with no rub at all.

11

Screw the pulley brackets to the top rail

KI 13



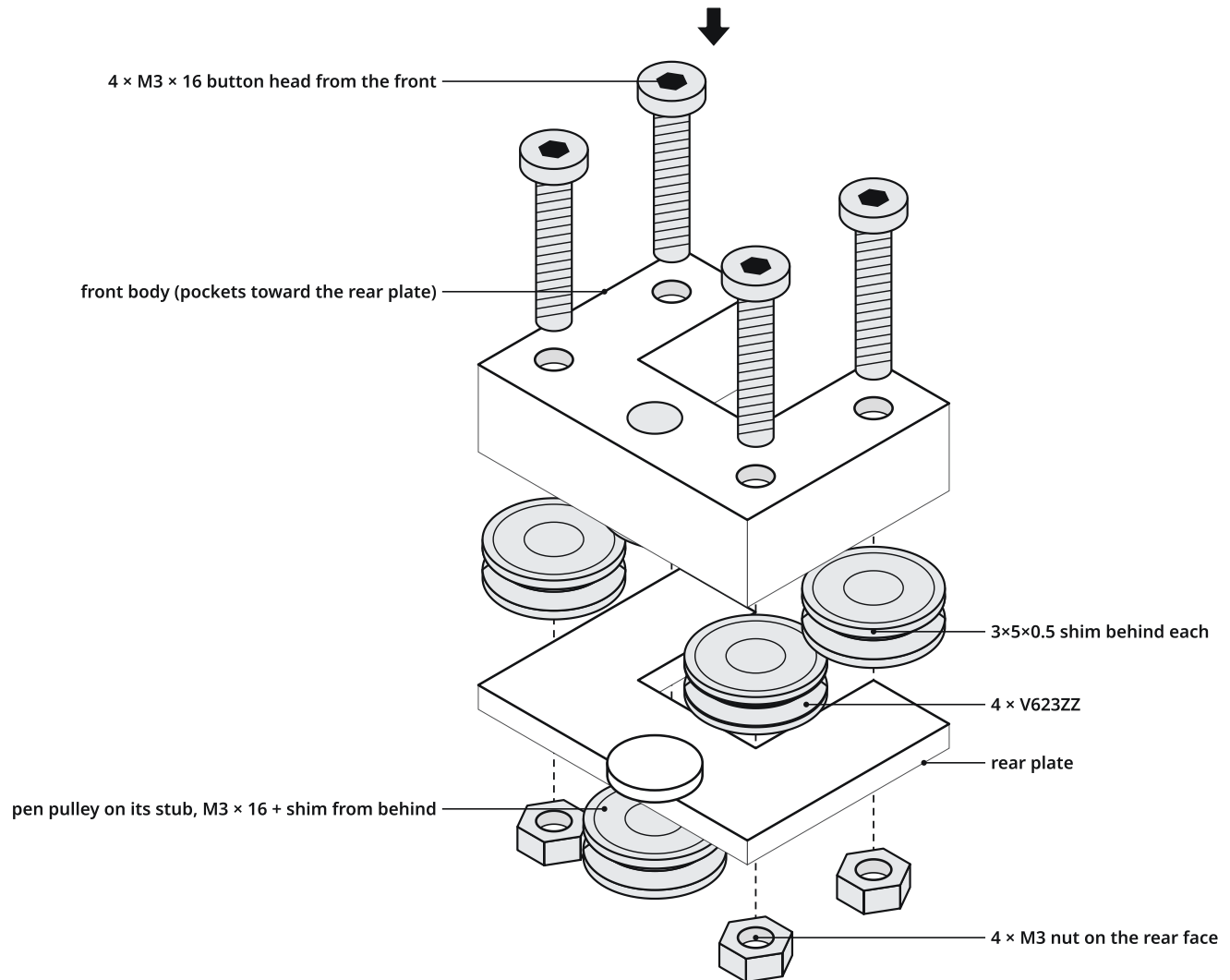
- Each bracket: two **V623ZZ** with a **3 × 5 × 0.5 shim** between them on an **M3 × 20**, head to the front, nyloc behind. Snug: they must spin freely.
- Five brackets directly above the shafts ($x - 224 \dots 0$) and the pen bracket at $x + 70$. **#4 × 3/4"** into pilots at station $x \pm 11, 15$ from the rail's rear edge; the notch on each bracket marks its station and faces front.
- The pen bracket carries one pulley (rear) and the screw-adjustable anchor (front).

i The two pulleys run in planes 4.5 apart, either side of the arm-bearing groove at $z - 15$. That's what lets the threading mirror at alternate stations (step 16).

12 Build the pen carriage

KI 6

KI 7



- Pen pulley on the rear plate's stub: **M3 x 16** with a shim, from behind, into the front body's insert.
- Four bearings, each with a shim on its rear side, clamped between rear plate and front body by **M3 x 16** from the front and a nut behind.
- Leave the +x pair (slotted holes) loose until the carriage is on its rods (step 13).

i The inner races are the spacers. The V grooves locate the carriage front to back; the two sides pushing outward on opposite rods locate it sideways.

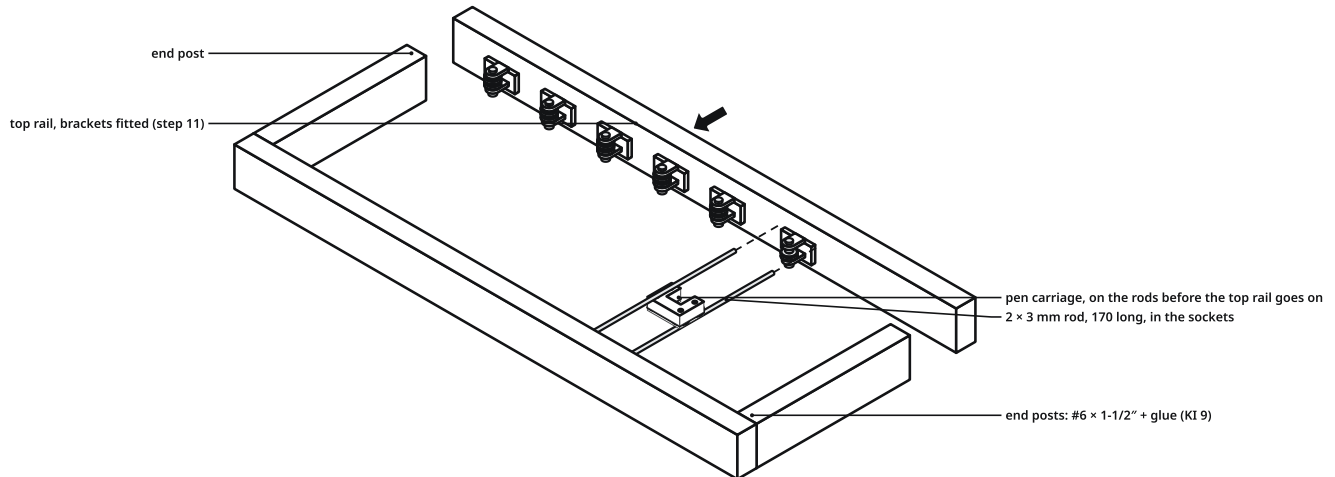
13

Close the frame around the rods and the carriage

KI 1

KI 9

KI 13



- Tape the 1:1 templates to the rails (**KI 13**) and drill them as a mirrored pair, clamped outer face to outer face: rod sockets $\varnothing 3.1 \times 9$ at 325.08 and 362.92 from the S2 end, 23.75 from the rear edge; the drum hole at 429.45, 24 from the rear edge.
- Bottom rail flat, both rods in its sockets, the carriage between them with its bearings on the rods. Then the top rail onto the rod ends.
- End posts between the rails, $\#6 \times 1-1/2"$ and glue. Push the +x bearing pair outward for a light preload and lock the two screws.

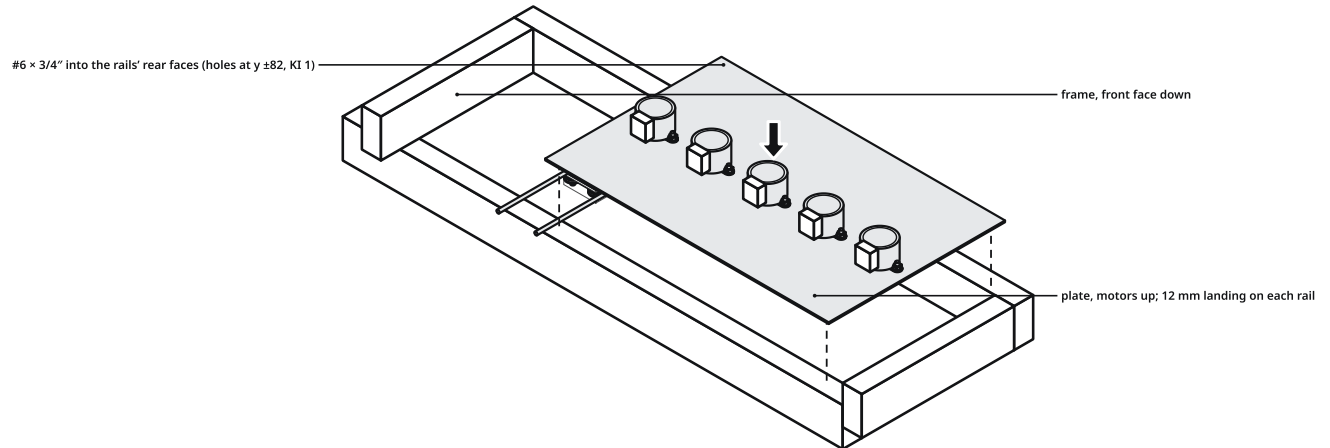


The rods are captive once the top rail is on: the carriage must go on them first.

14

Screw the faceplate to the frame

KI 1

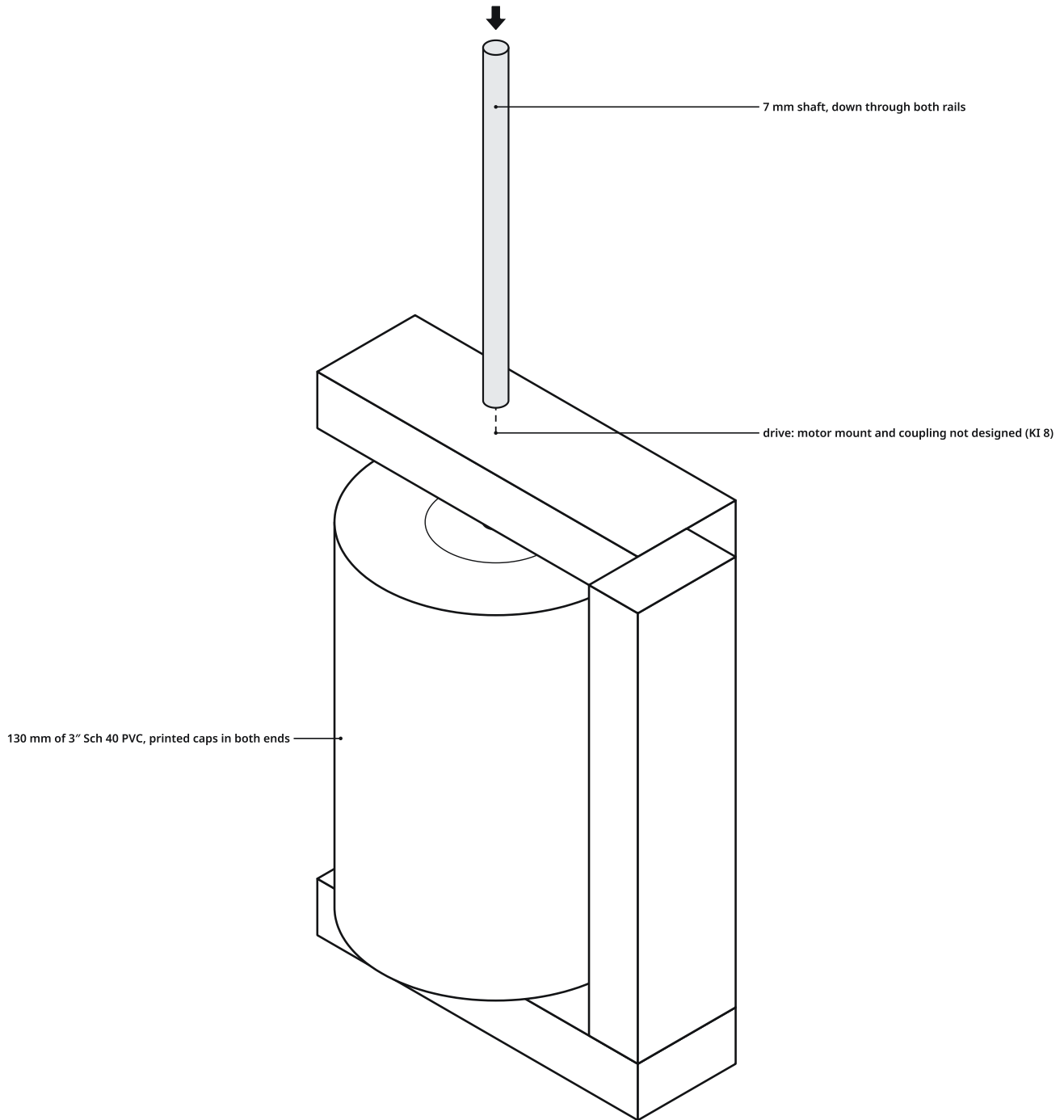


- Frame face down. The plate comes down onto the rails' rear faces, motors up, with **12 mm of plate landing on each rail**.
- **#6 × 3/4"** through the Ø4.0 holes into the rails. Then stand the machine up and check every arm bearing clears the pulleys above it (18 mm at M2).

15

Fit the drum

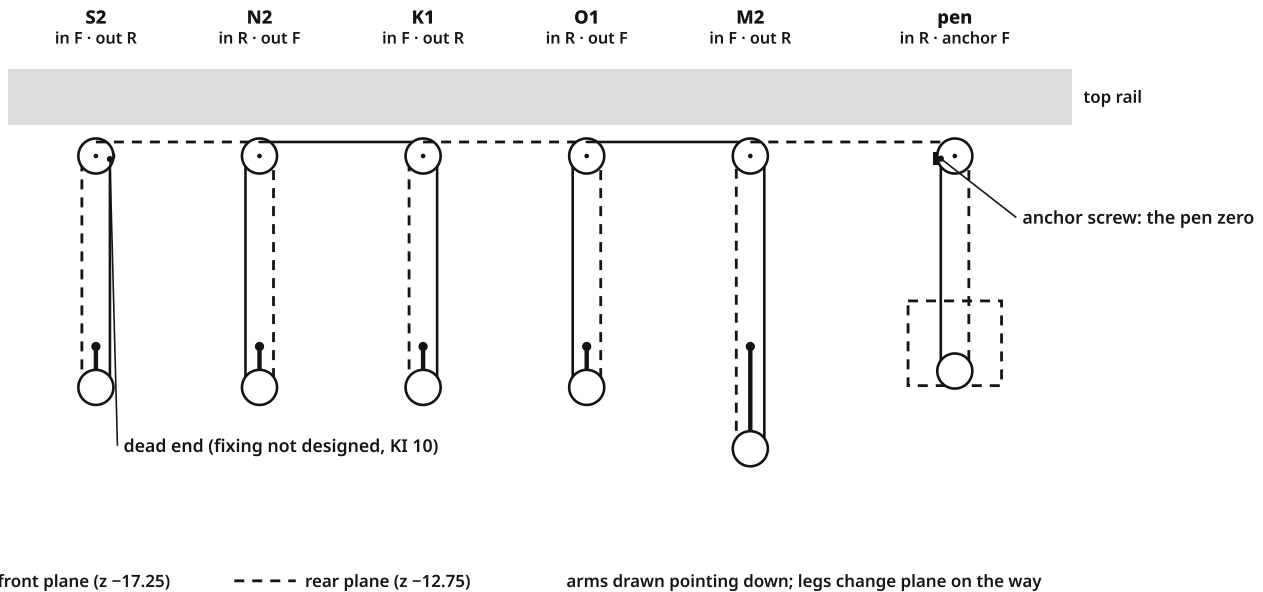
KI 8



- Cut 130 mm of 3" PVC square; press an end cap into each end.
- The 7 mm shaft down through the top rail, the drum and into the bottom rail's plain bearing, with a washer under the bottom cap.
- The drum axis sits at x 155.5, z -24, on the carriage's mid-plane; 5.95 mm clear of the right post.

16 Thread the line

KI 10



	S2	N2	K1	O1	M2	pen
in over	front	rear	front	rear	front	rear
out over	rear	front	rear	front	rear	front = anchor

- Crimp a loop in the far end of the **0.30 mm fluorocarbon** and fix it at S2's bracket, front plane.
- At each station: over the IN pulley, down the +x side, **round under the arm's bearing**, up the -x side, over the OUT pulley. Then along the rail to the next one.
- At the pen: over the rear pulley, round under the carriage's pen pulley, up to the anchor screw on the front plane. Crimp it there.

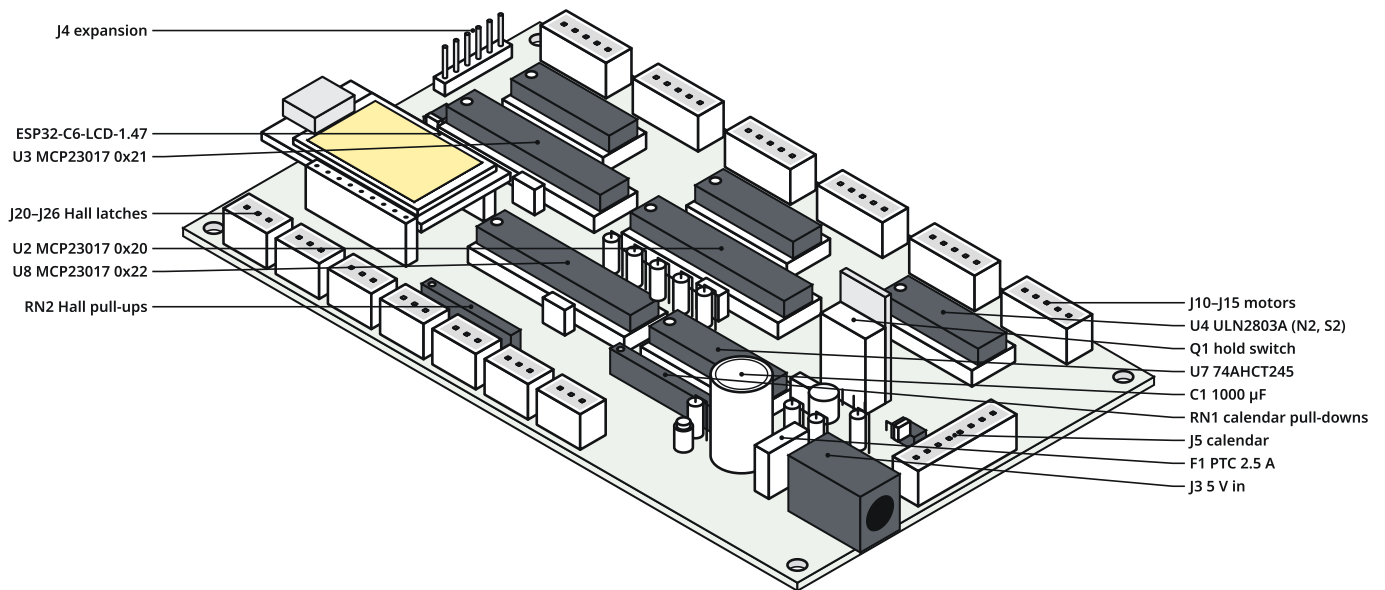
i The order mirrors at every station, so every run between stations stays in one plane and only the legs skew (about 2°). The line passes **round** each arm bearing and never attaches: the U-turn turns a rise d into $2d$ of take-up, and the pen pulley halves it again, so the pen moves 35 mm per metre of tide.

B

Main board

140 × 81.3 mm, all through-hole, chips and the ESP32 in sockets. Solder in rounds, lowest parts first; test where a round says so.

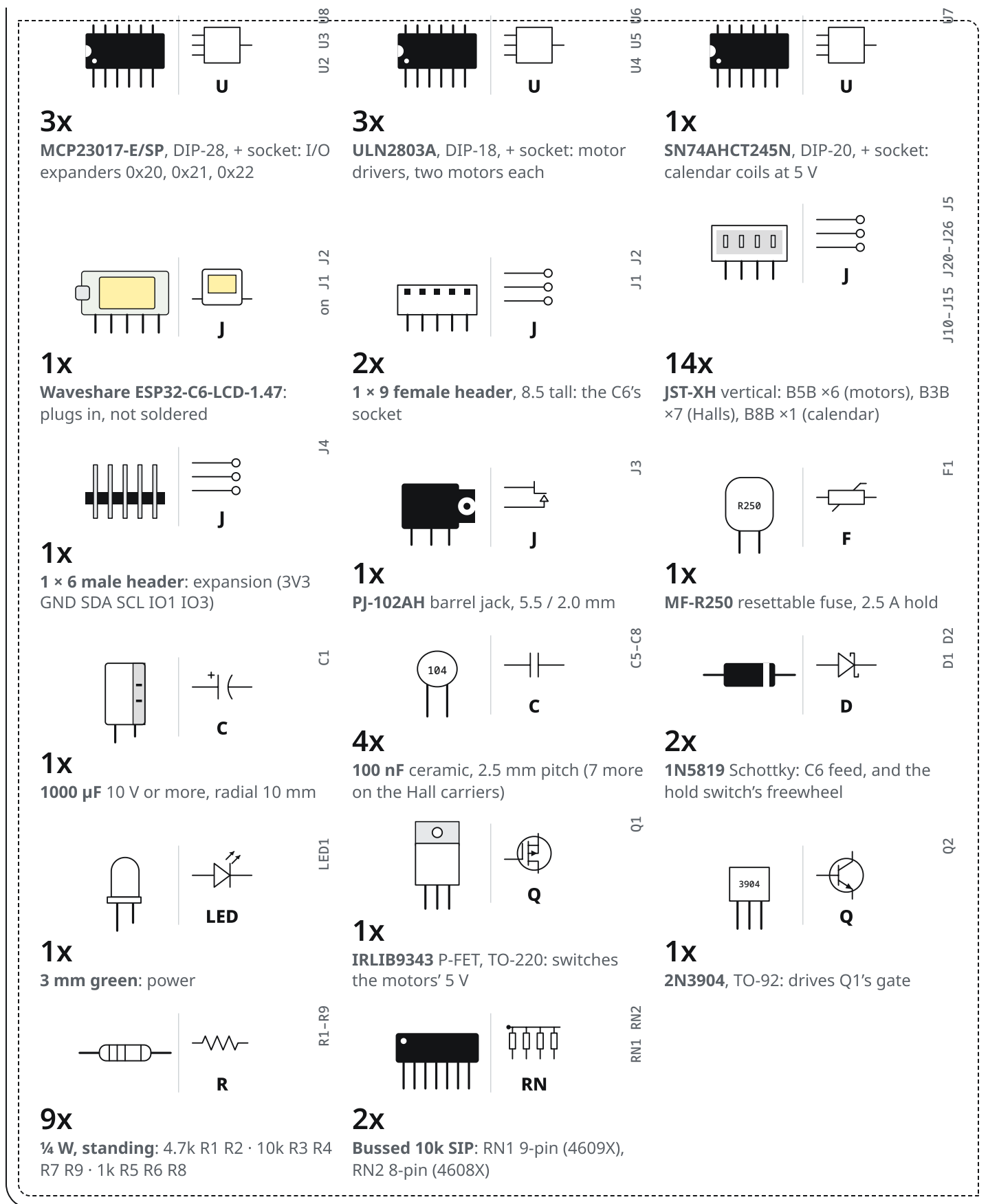
The finished board



Motor sockets across the top edge, Hall sockets along the bottom, calendar and power on the right, the ESP32-C6-LCD-1.47 on the left with its USB-C out of the left edge. Nothing stands under the C6.

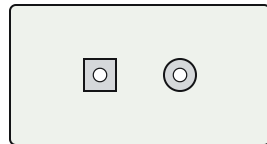
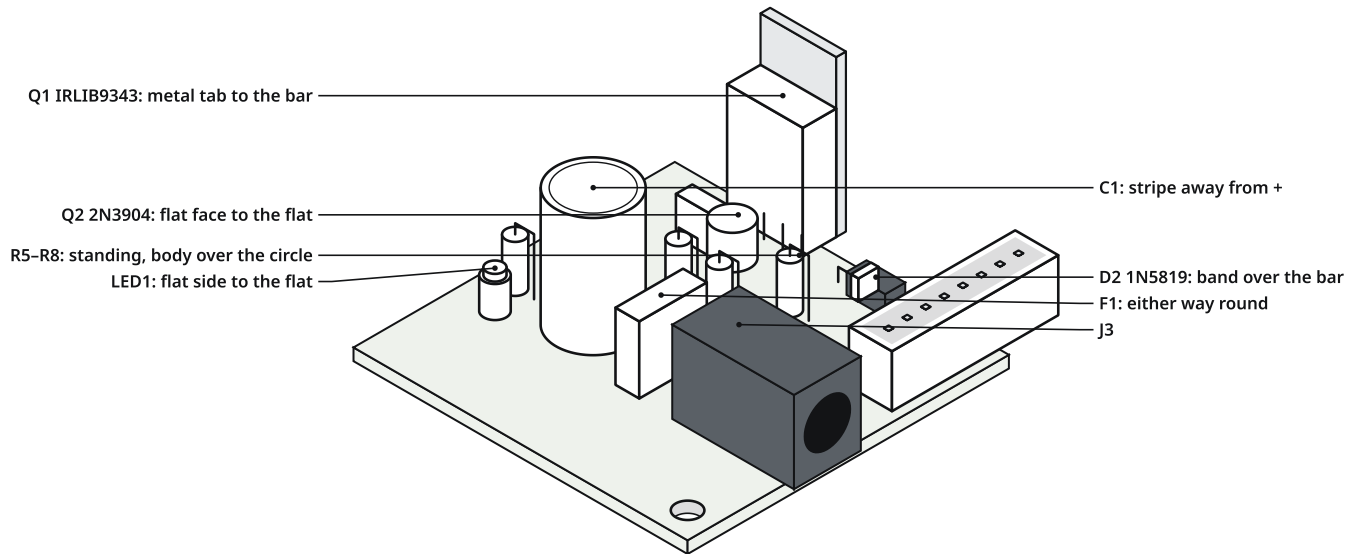
4

Electronic parts for the board

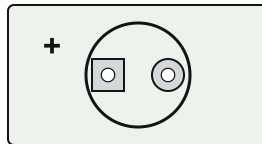


17

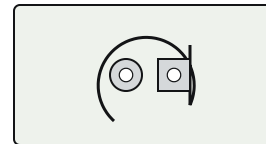
Read the silkscreen



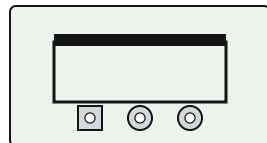
square pad = pin 1, +, or K



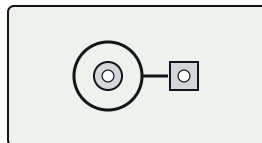
"+" = long leg



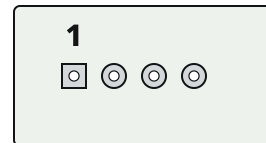
flat edge = flat of the part



bold bar = metal tab side



circle = where the body stands

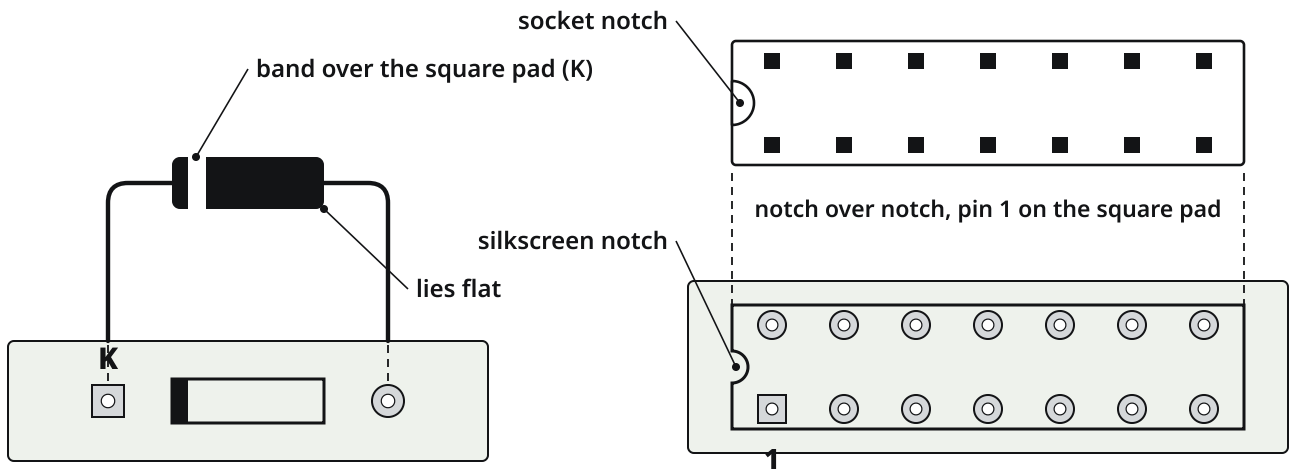


"1" marks pin 1 of a socket

Every part on this board has one right way round except the resistors, the ceramic capacitors and F1. The marks above are the whole vocabulary.

18

Round 1: diodes and IC sockets

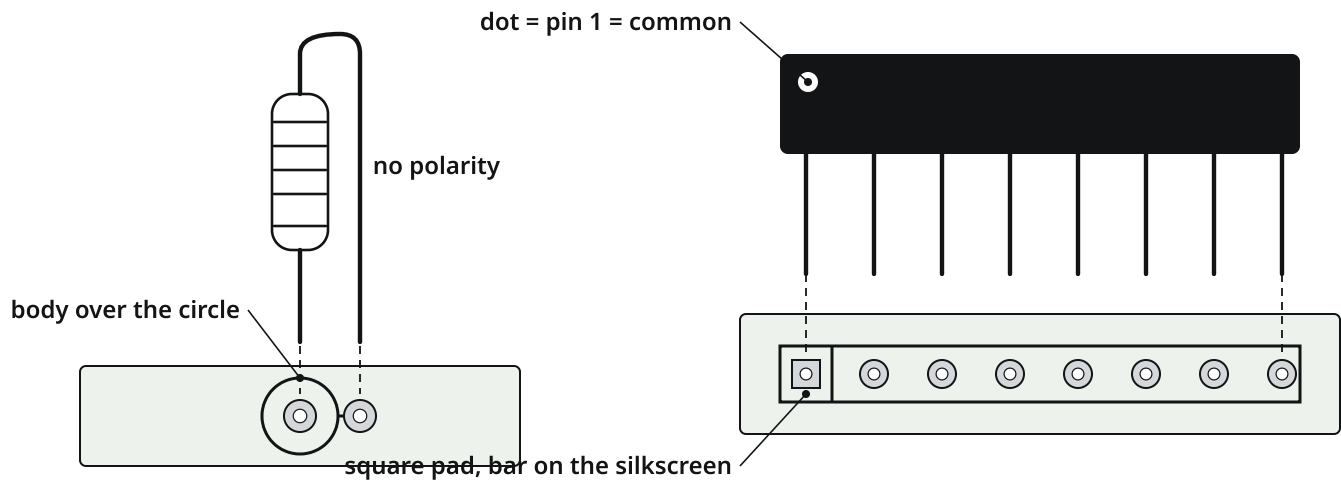


- **D1, D2** (1N5819) lying flat, band over the square pad.
- **Sockets** for U2, U3, U8 (28 pin), U4, U5, U6 (18 pin), U7 (20 pin): notch over the silkscreen notch. Solder two opposite corners, check it sits flat, then the rest.

No chips yet: they go in at Test 6.

19

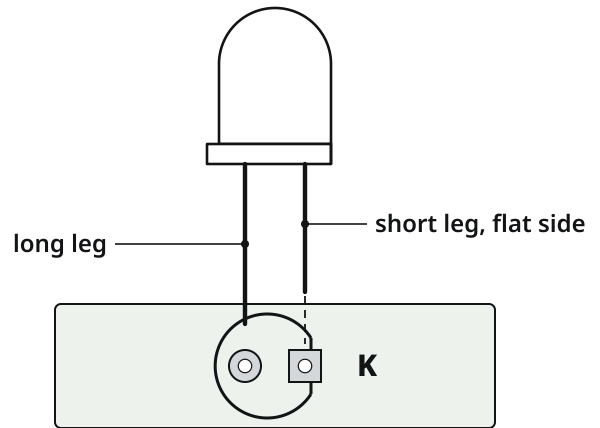
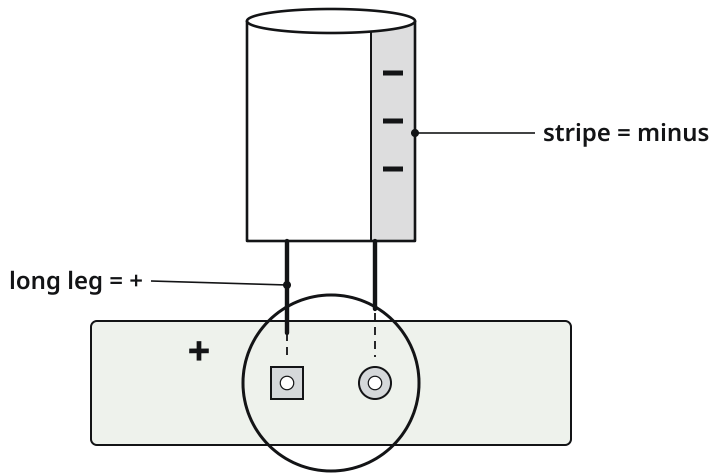
Round 2: resistors, networks, small capacitors



- **R1–R9** standing, body over the circle: 4.7k R1 R2; 10k R3 R4 R7 R9; 1k R5 R6 R8.
- **RN1** (9 pins) and **RN2** (8 pins): the dot is pin 1, the common, on the square pad.
- **C5–C8**, 100 nF, either way round.

20

Round 3: power entry

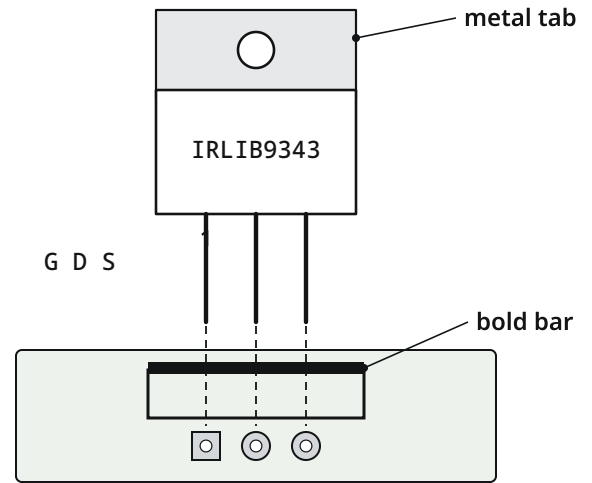
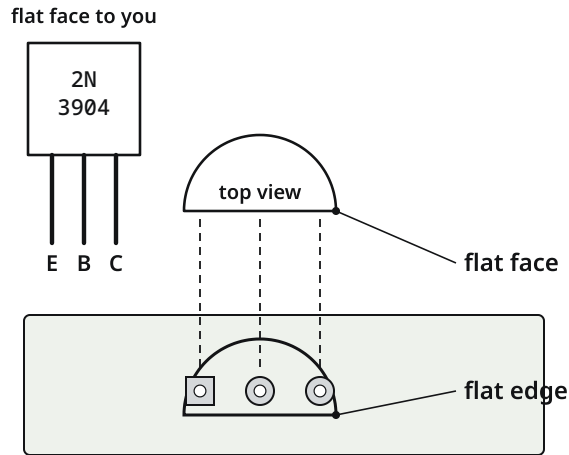


- **F1** (resettable fuse), either way round. **C1** 1000 μ F: long leg in the + pad. **LED1**: flat side on the flat, short leg in the square pad.
- **J3**, the barrel jack: it overhangs the right edge by design.

Now do Tests 0, 1 and 2 before going on.

21

Round 4: the hold switch



- **Q2** (2N3904): flat face on the flat edge.
- **Q1** (IRLB9343): metal tab to the bold bar, standing upright, no heatsink.

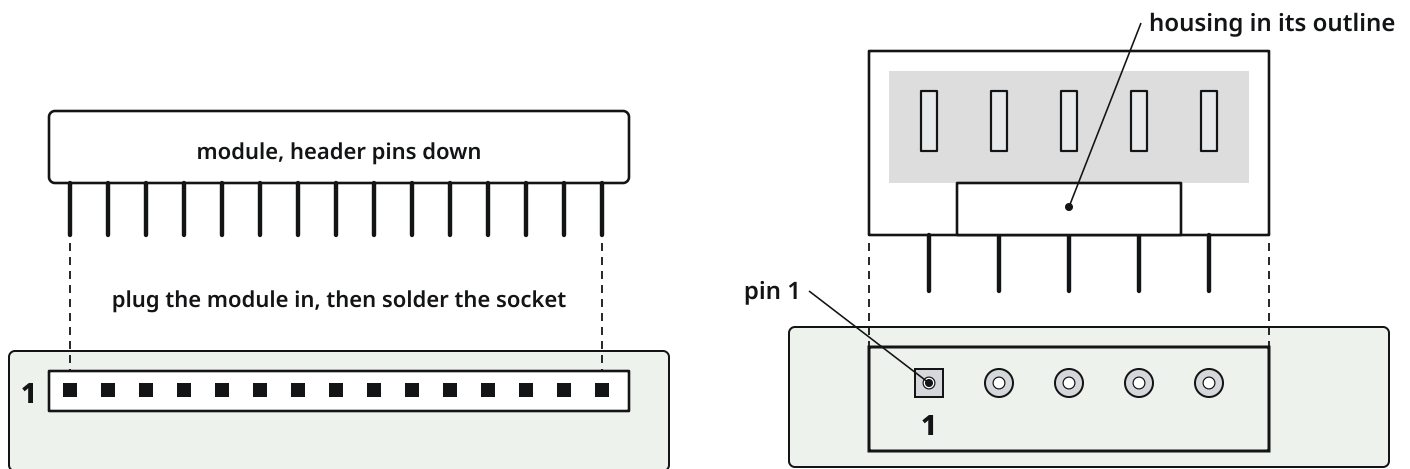


Q1 switches the motors' 5 V (the HOLD rail) so the firmware can drop the holding current to 60% between steps. Fit it anyway; step 39 decides whether the hold is needed at all.

Then Test 3.

22

Round 5: connectors



- **J1, J2:** plug both headers onto the C6's pins, set it in the board and solder; then **unplug the C6** again.
- **J10-J15** (5 pin), **J20-J26** (3 pin), **J5** (8 pin): housing inside its outline, pin 1 on the square pad.
- **J4**, the 1 × 6 expansion header.

Then Tests 4 and 5.

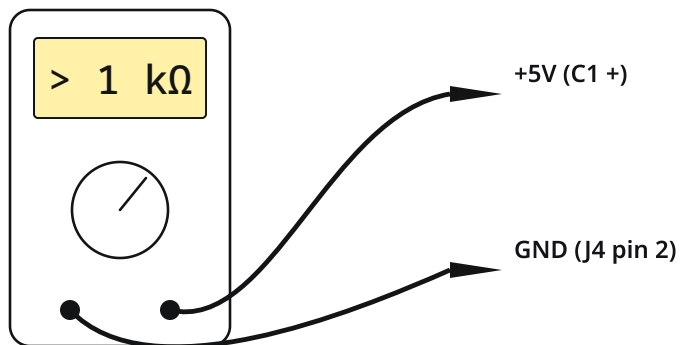
C

Smoke test

Nothing expensive goes in until every rail reads right. Use a bench supply at 5.0 V with a 100 mA limit, or a 5 V supply through a USB power meter.

23

Test 0: Look, then measure



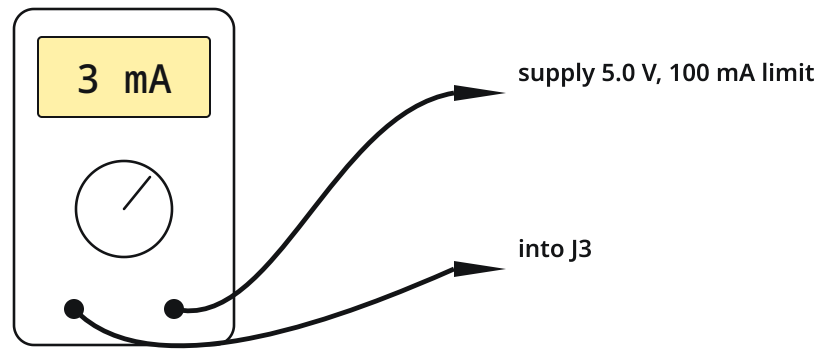
- Look at every joint under a lens: no bridges, no dull balls. Socket notches, diode bands, C1's stripe, Q1's tab: all as the silkscreen.
- Unpowered, meter on ohms: **+5V to GND** climbs as C1 charges and ends above 1 kΩ. Same from **J3's centre pin** to GND.



A reading that stays near 0 Ω is a short: find it before any power goes on.

24

Test 1: First power, current-limited



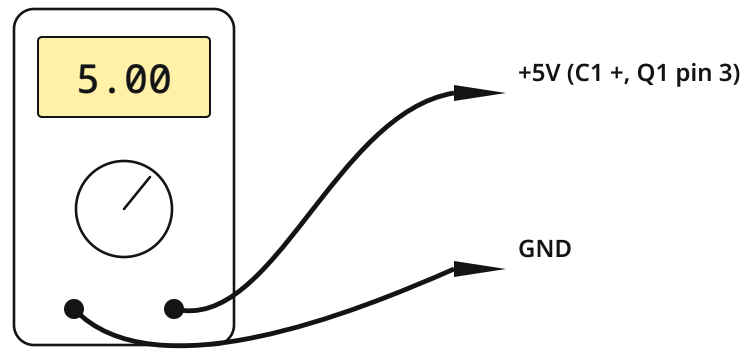
- Power into J3. **LED1 lights** and the supply reads about **3 mA**: (5.0 – 2.1 V) through R8's 1 k Ω .



If the supply hits its limit, switch off at once and go back to Test 0.

25

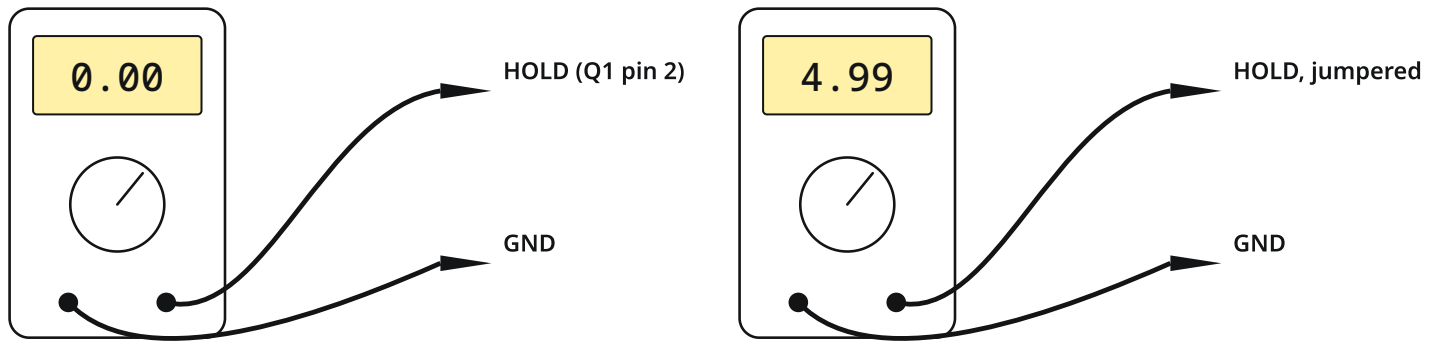
Test 2: The 5 V rail



- 4.95 to 5.00 V: the supply less F1's drop at 3 mA, which is next to nothing.

26

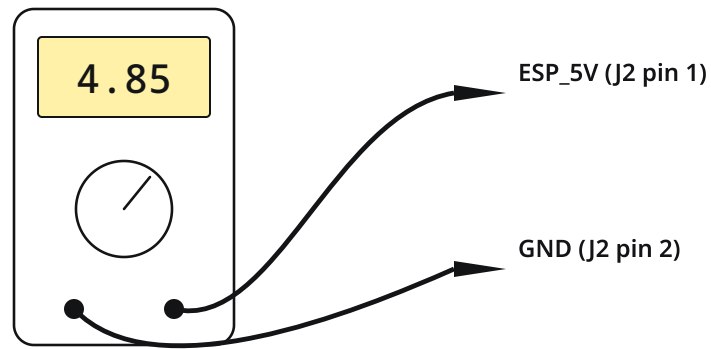
Test 3: The hold switch



- Power on, nothing plugged in: **HOLD reads 0 V**. R7 holds Q2 off, so R5 holds Q1 off.
- Jumper **J2 pin 6 (PWM) to J2 pin 1 (ESP_5V)**: HOLD rises to **about 4.99 V**. Remove the jumper.

27

Test 4: The ESP32's 5 V

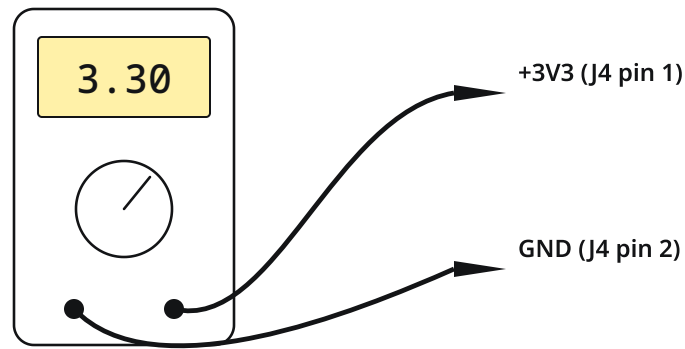


- 4.8 to 4.95 V: +5V less D1's Schottky drop at almost no load.

i D1 keeps the supply and the C6's own USB apart, so flashing over USB with the board powered is safe.

28

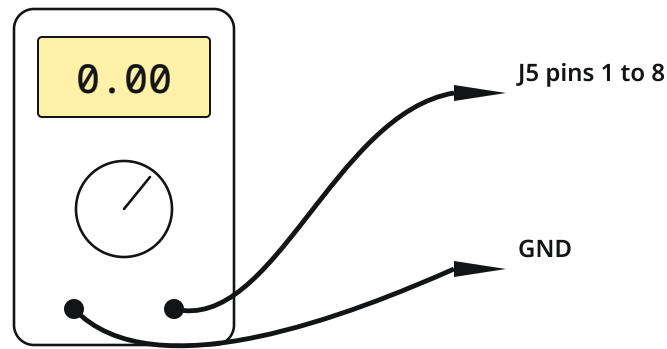
Test 5: Plug in the C6



- Power off. Plug the C6 into J1/J2, LCD up, USB-C at the board's left edge. Power on.
- **3.30 V** on J4 pin 1 (from the C6's regulator). Also 3.3 V on SDA and SCL (J4 pins 3, 4; R1, R2), on INT_A and INT_B (J1 pins 6, 5), and on pin 2 of every Hall socket J20–J26 (RN2).

29

Test 6: Chips in



- Power off. Chips in, notch to notch: **U2, U3, U8** MCP23017, **U4, U5, U6** ULN2803A, **U7** 74AHCT245. Check no leg folded under.
- Power on. Every calendar line on **J5 reads 0 V** (RN1 holds U7's inputs low until the 0x22 expander is set up), and HOLD still reads 0.
- The current is the C6's own draw plus about 3 mA. Above 250 mA: switch off and look for a chip in backwards.

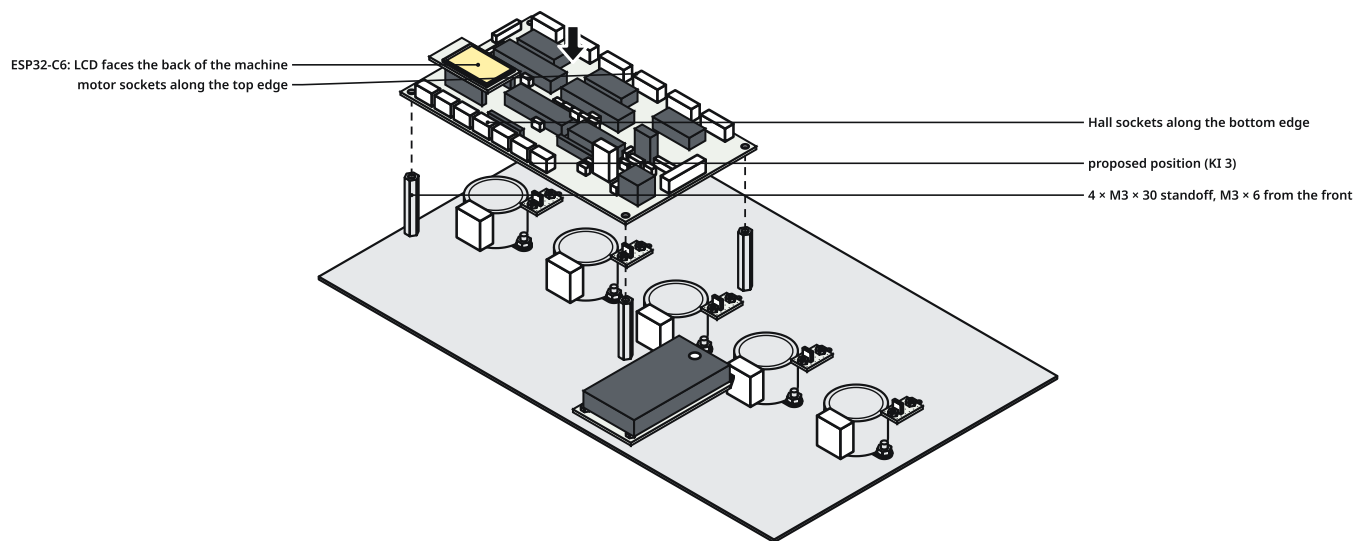
The I²C scan (0x20, 0x21, 0x22) is step 38: it needs firmware.

D Wiring

The board goes on the back of the plate, 30 mm off it, clear of the motors. Everything plugs into its edges.

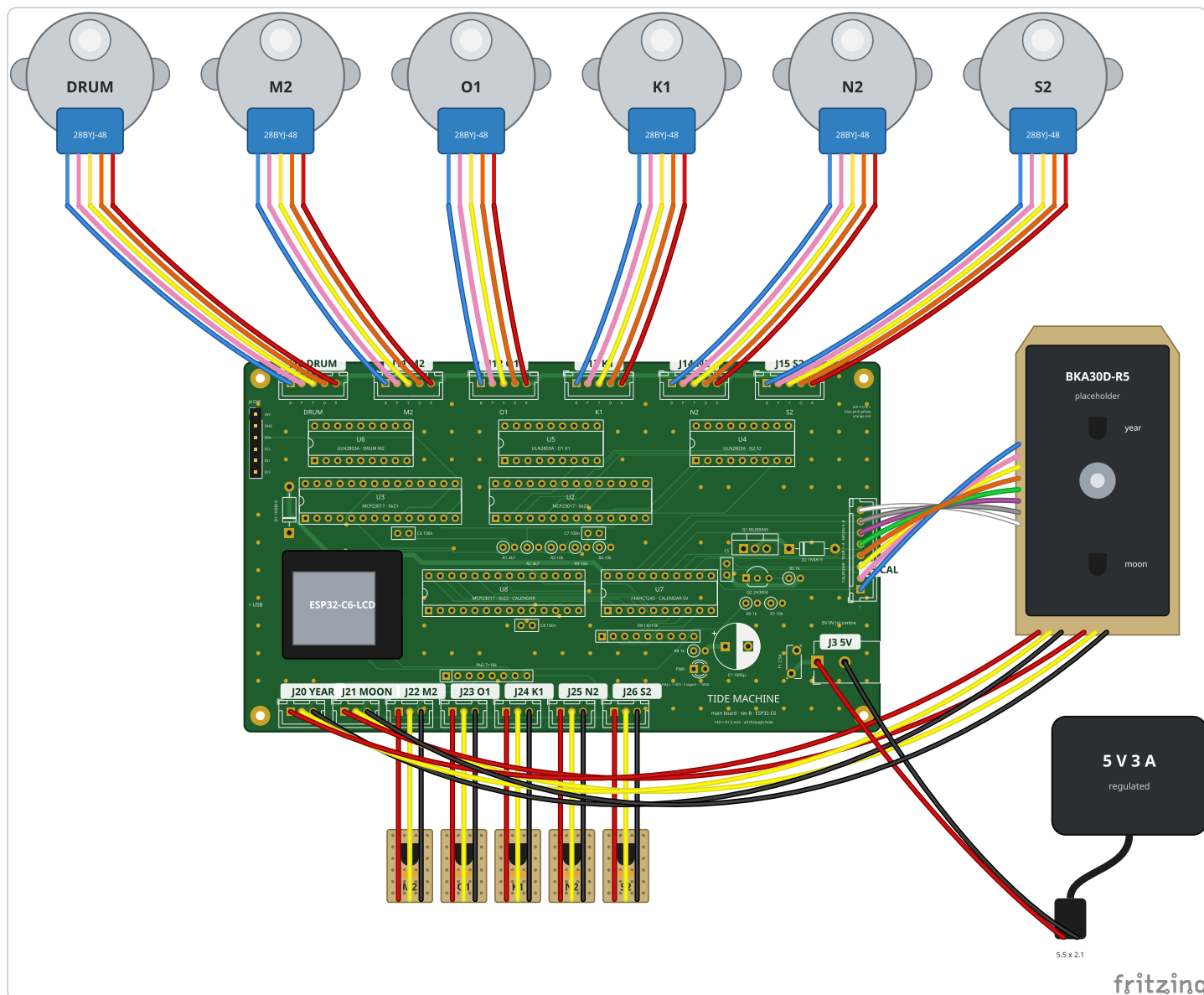
30 Mount the board on its standoffs

KI 3



- Four **M3 x 30** standoffs, **M3 x 6** from the front of the plate. The board on top, component side out, **silkscreen reading upright from behind**, four more **M3 x 6**.
- The board is 30 mm off the plate, the motors 20.5 deep: 8 mm under the lead ends.

31 The whole loom

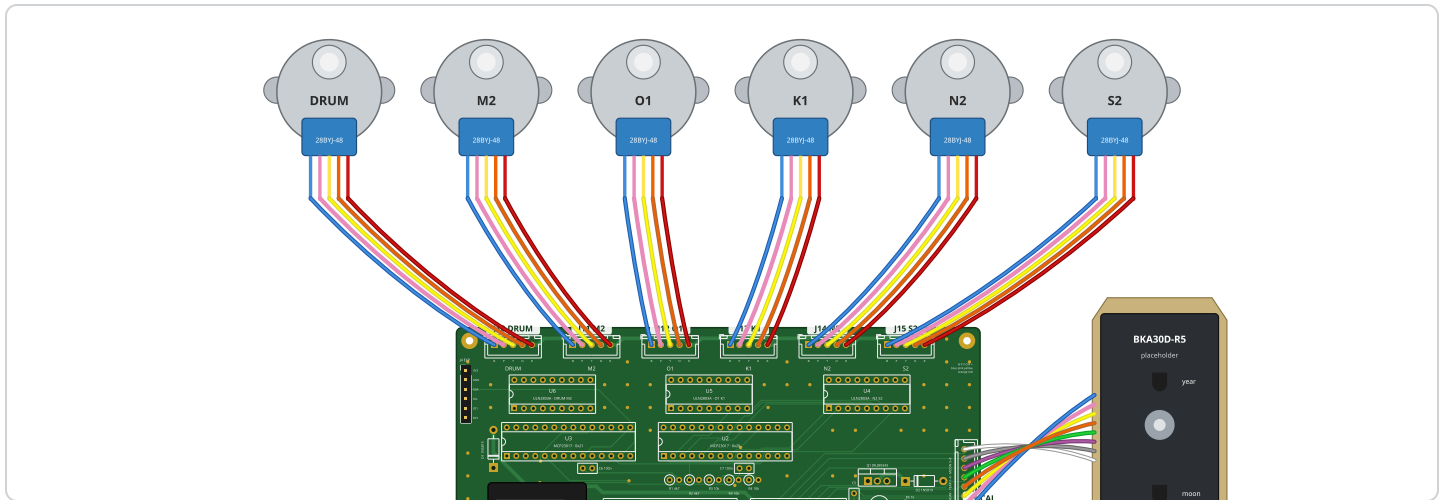


Fritzing, breadboard view. Source: `docs/assembly_guide/tide_wiring.fzz`, generated by `gen_fritzing.py`; the board is drawn from `pcb/tide_main.kicad_pcb`.

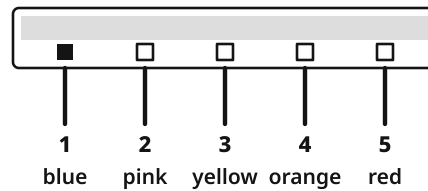
32

Plug in the motors

KI 8



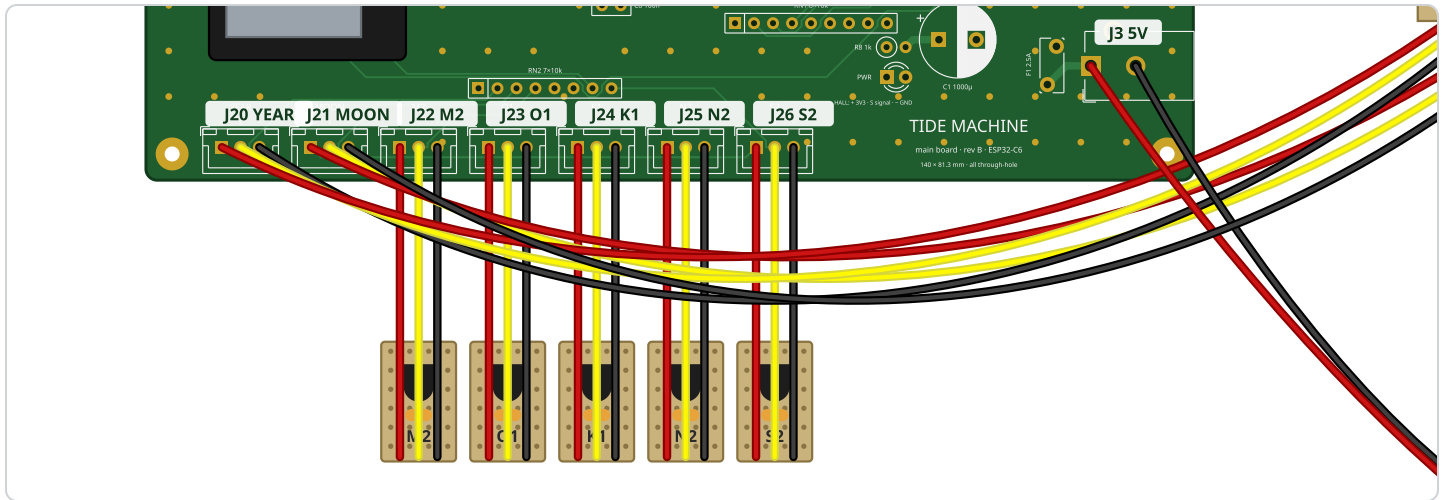
28BYJ-48 plug into J10-J15



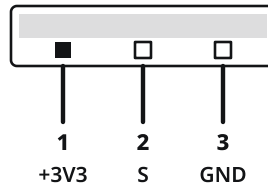
- Each 28BYJ-48's own plug into its socket: **J10 drum**, **J11 M2**, **J12 O1**, **J13 K1**, **J14 N2**, **J15 S2**, as the silkscreen says.
- The plug is keyed. Red is pin 5, the switched HOLD rail.

33

Crimp and plug the Hall leads



XHP-3 into J20-J26

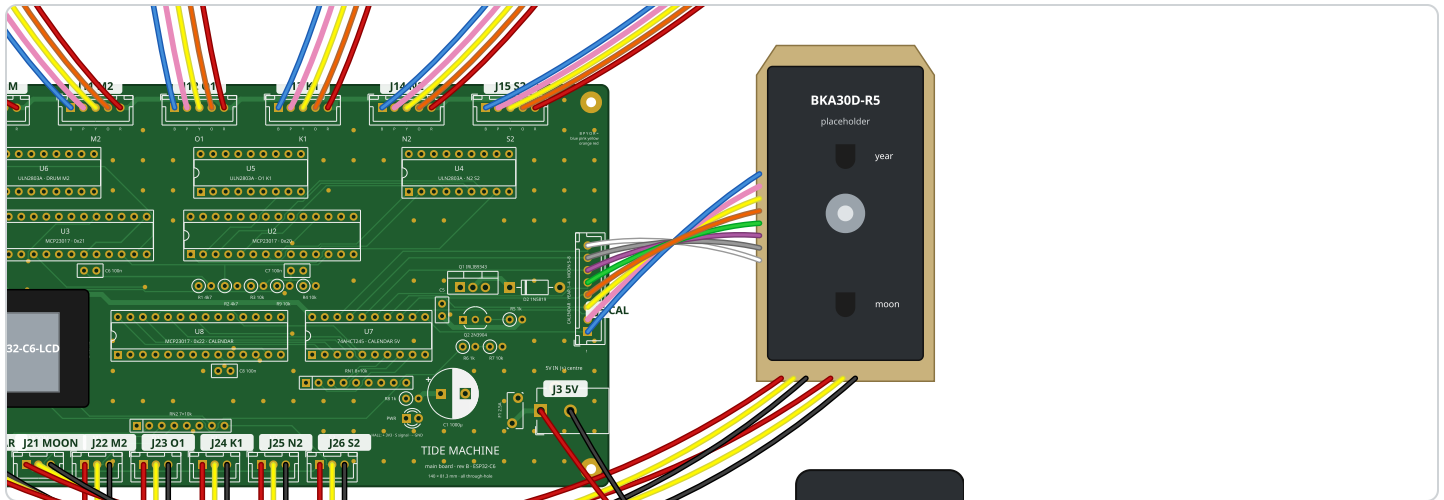


- Each 3-core lead into an **XHP-3** : pin 1 **+3V3**, pin 2 **S** (the latch output), pin 3 **GND**.
- **J20 year, J21 moon, J22 M2, J23 O1, J24 K1, J25 N2, J26 S2.**

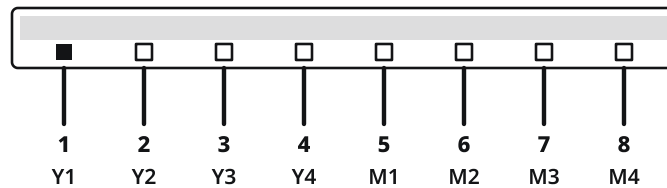
i The pull-ups are on the board (RN2, 10 kΩ to 3.3 V), so each carrier needs only its latch and a 100 nF.

34 The calendar lead

KI 5



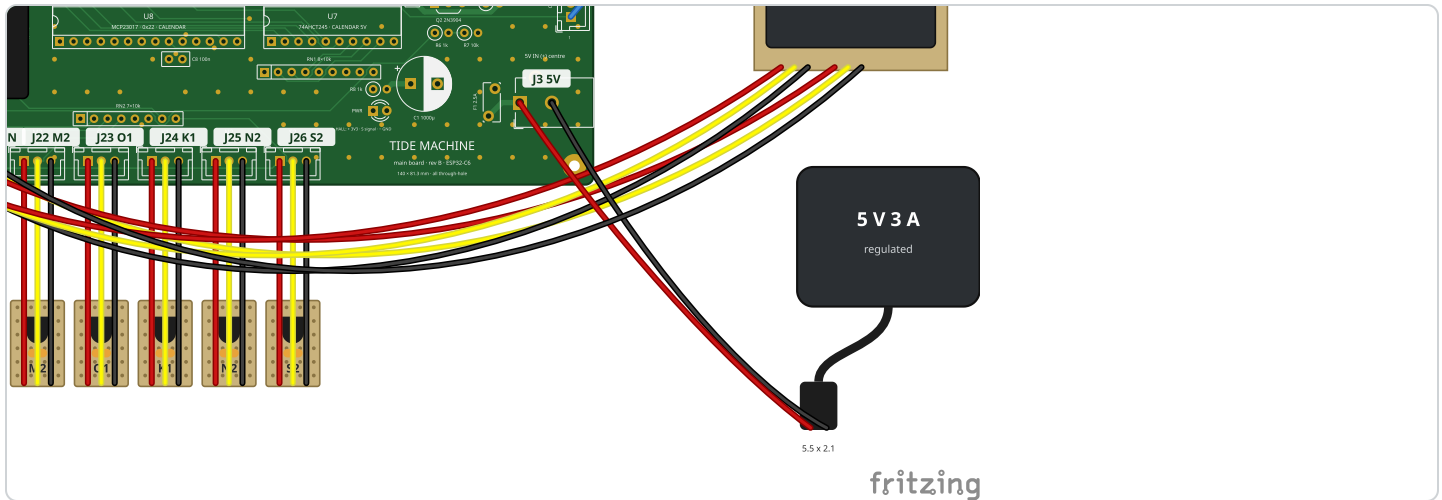
XHP-8 into J5 (year coils 1-4, moon 5-8)



- 8-core from the BKA30D-R5 into an XHP-8 on J5: pins 1-4 the year (outer) motor, pins 5-8 the moon (inner) motor.
- Match each motor's coil pins to Y1-Y4 and M1-M4 from its datasheet when it's in hand.

35

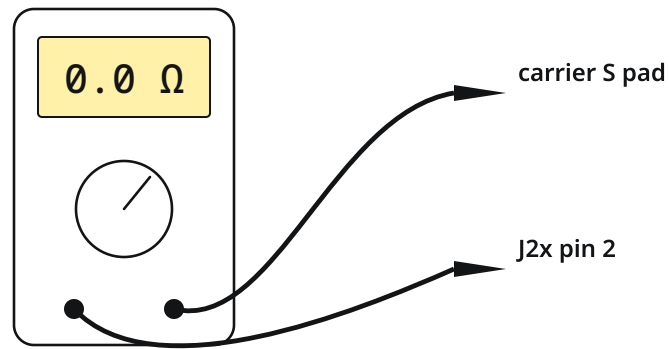
Power and USB



- A **5 V 3 A regulated** supply, 5.5 × 2.1 plug, centre positive, into J3. Peak draw is about 1.5 A with every motor energised.
- USB-C on the C6, at the board's left edge, for flashing and the serial console.

36

Pin check before power



- Power off. Beep every Hall lead end to end (+, S, -) and the calendar lead pin by pin.
- +3V3 to GND on each Hall socket must **not** beep.

E

First power-up and calibration

Each step is tagged **today** (works with what's in the repo now) or **planned** (needs firmware that isn't written yet).

37

Flash the firmware

planned

- Over the C6's USB-C. The firmware carries the tables from `pcb/design.py` : `MOTOR_BITS` , `HALL_BITS` , `CAL_BITS` , `CAL_HALL_BITS` , `ESP_GPIO` , plus a per-station inverse table from `kinematics.py` .

 The C6's LCD faces the back of the machine: planned as a status screen (time, phase of each station, last Hall edge).

38

I²C scan

planned

- Expect exactly **0x20, 0x21, 0x22** on SDA GPIO19 / SCL GPIO18. A missing address is a chip not seated or an address pin: U2 all low, U3 A0 high, U8 A1 high.

39

Holding-torque test

today

- One spare 28BYJ-48 on the bench, a 100 mm lever on its shaft, a known weight on the end: torque = $m \cdot g \cdot 0.1$.
- Unpowered first, then with one coil energised (red to 5 V, orange to GND).

 M2's arm needs 14.5 mN·m. The datasheet's 59–118 mN·m “friction torque” hints the gearbox holds that unpowered. If so, the PWM hold can go to zero between steps.

40

Check every motor's direction

planned

- At 1000×, one station at a time: each turns the way its table says. Every shaft turns one way only, which is what keeps gearbox backlash a constant.

41

Hall edges

planned

- Each station's latch reports one edge per turn through the 0x21 expander's INTB (GPIO20); year and moon through 0x22 (GPIO23).
- Run each station past its edge ten times: the step count between edges must repeat exactly (2048 per turn at the output shaft, less the table's shape).

42

Home the calendar

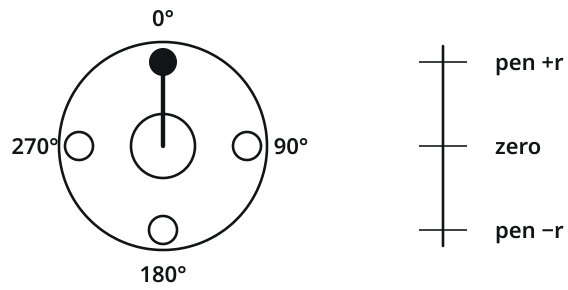
planned

- Slowly (the gauge motor is weak): year dial to Jan 1, moon disc to new moon, each on its latch edge. Then to today's date and the **true** moon phase.

43

Go-to-phase: check each arm's radius

planned



- One station at a time: command its phase to 0°, 90°, 180°, 270°. The pen moves $\pm r$ about its zero.

44

Set the pen zero

planned

- **Go-to-zero** (every output at 0), then turn the anchor screw on the pen bracket until the pen sits on the chart's datum line.

 The pen's zero is set by line length and nothing else. Homing can't crash the pen: its travel is bounded by $\Sigma r = 57.05$ against 62 mm available.

45

Soak at 1000×

planned

- 90 minutes at 1000× (60 days of tide) against the Marigram Sandbox. No skips: every Hall edge arrives on its predicted step.

F

How to use

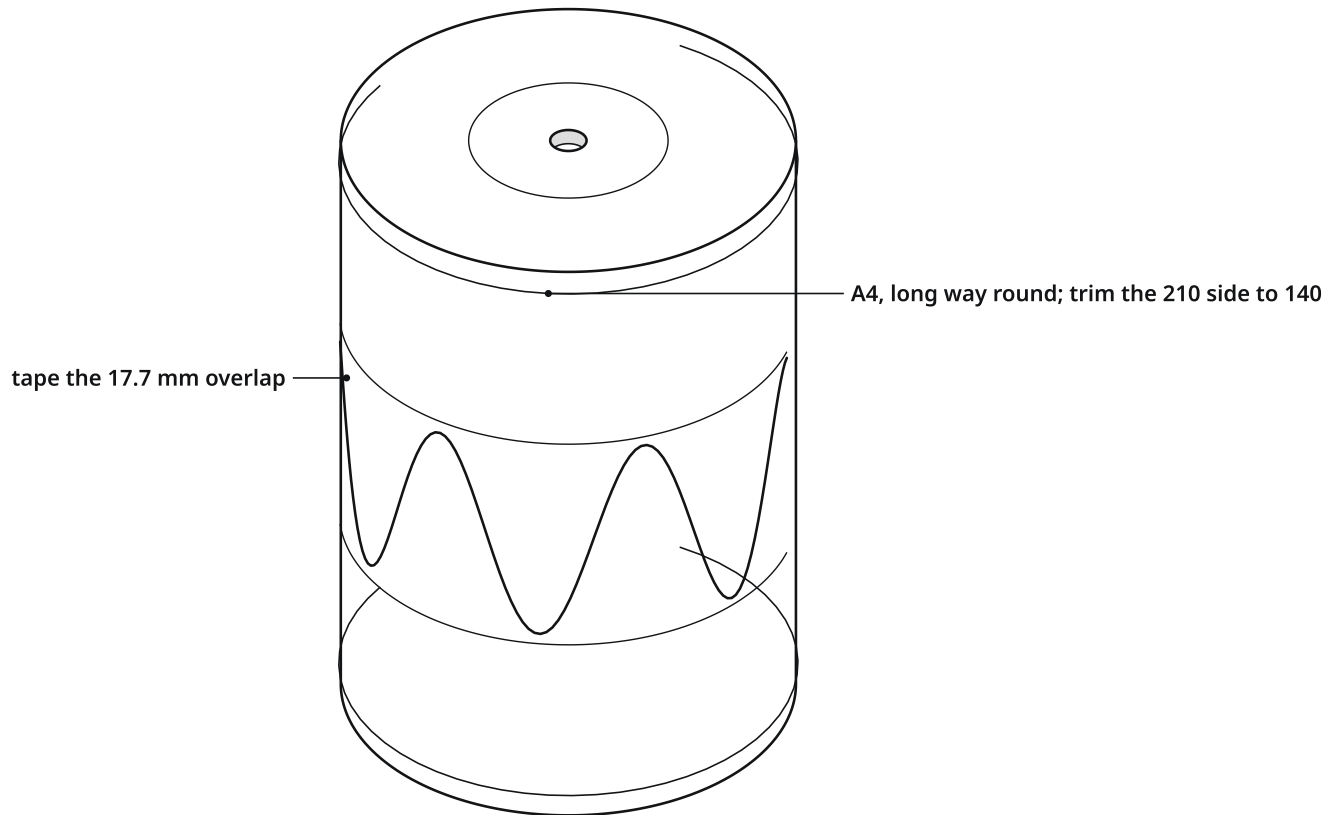
Set it running, load paper, read the dials.

46

Start it

planned

- Power on: Wi-Fi, NTP, re-home every station, then run in real time. NTP again daily.



- A4 wrapped the long way round the 279.3 mm drum; trim the 210 side to 140. Tape the 17.7 mm overlap.

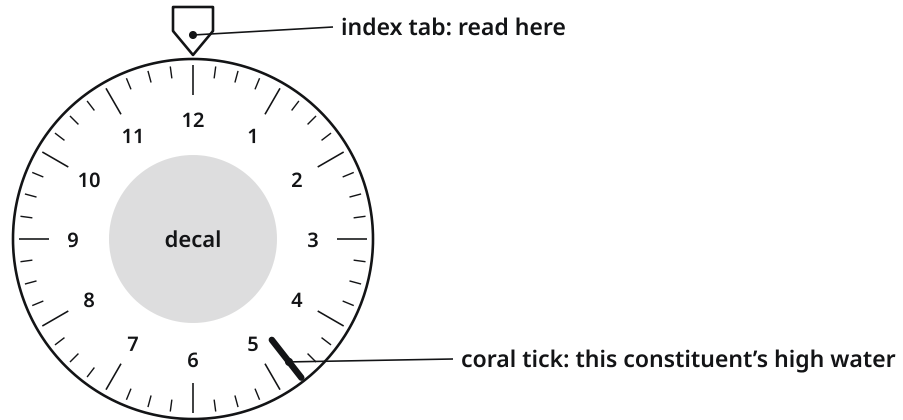
i Chart speed is still open: 7 days a turn (20.6 mm per tide) or 3.5 days (41.3 mm, a classic marigram). It's a firmware constant.

48

Replay at speed

planned

- The speed-up multiplier: 1000× replays 60 days in 86 minutes. Stay under 5000×, and re-home after any skip.



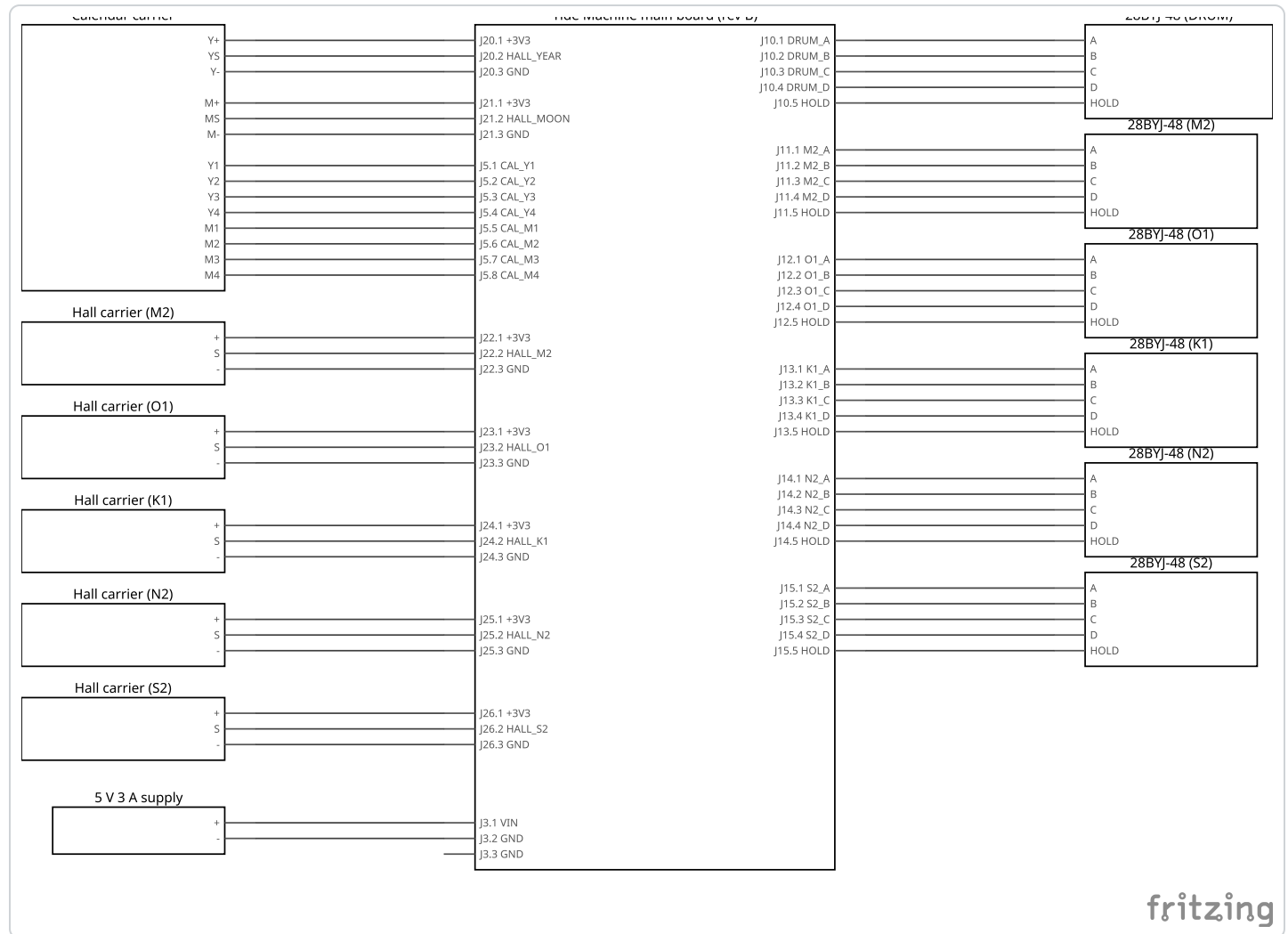
- Read each scale under its index tab; readings increase clockwise. **S2**: local mean solar time. **M2**: lunar hours. **K1**: sidereal time. **N2**, **O1**: hours of their own periods.
- The coral tick is each constituent's high water. On M2 the arm itself is the pointer.
- Calendar: the hand reads the month on the year dial and days since new moon on the moon disc.

G

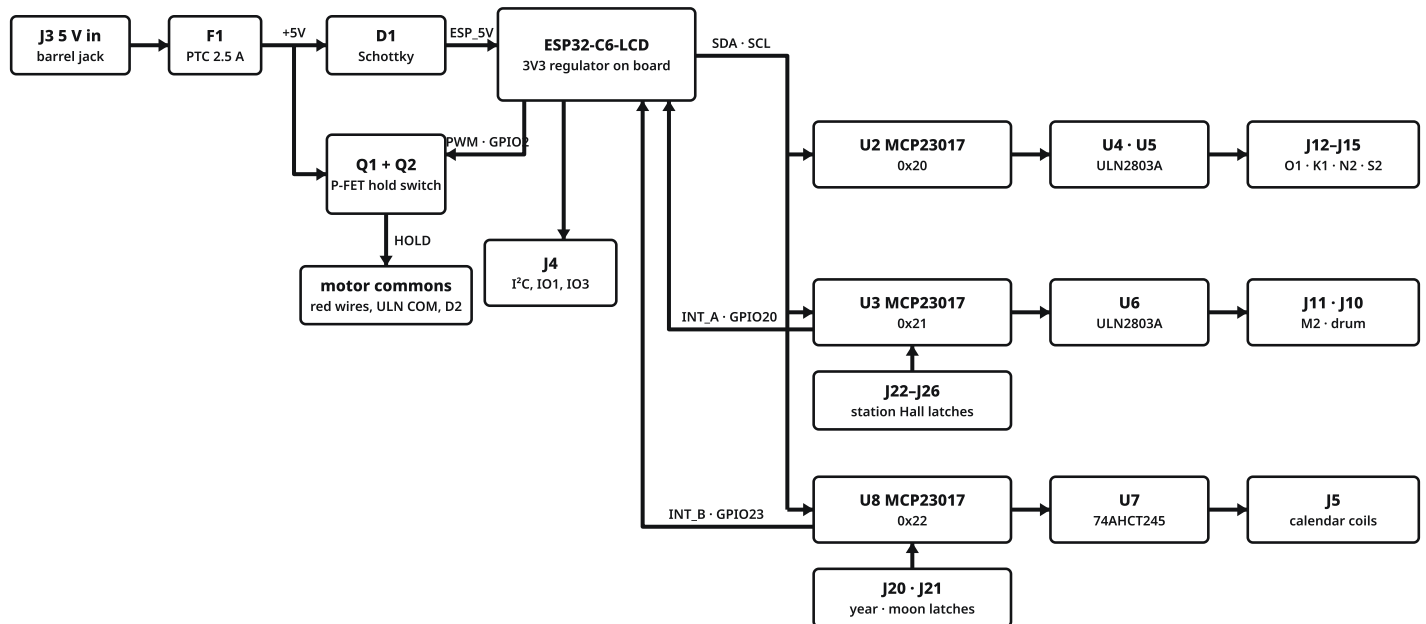
Appendix: schematics

The whole system, the board as blocks, then the board's full schematic.

G1 · System wiring, schematic view



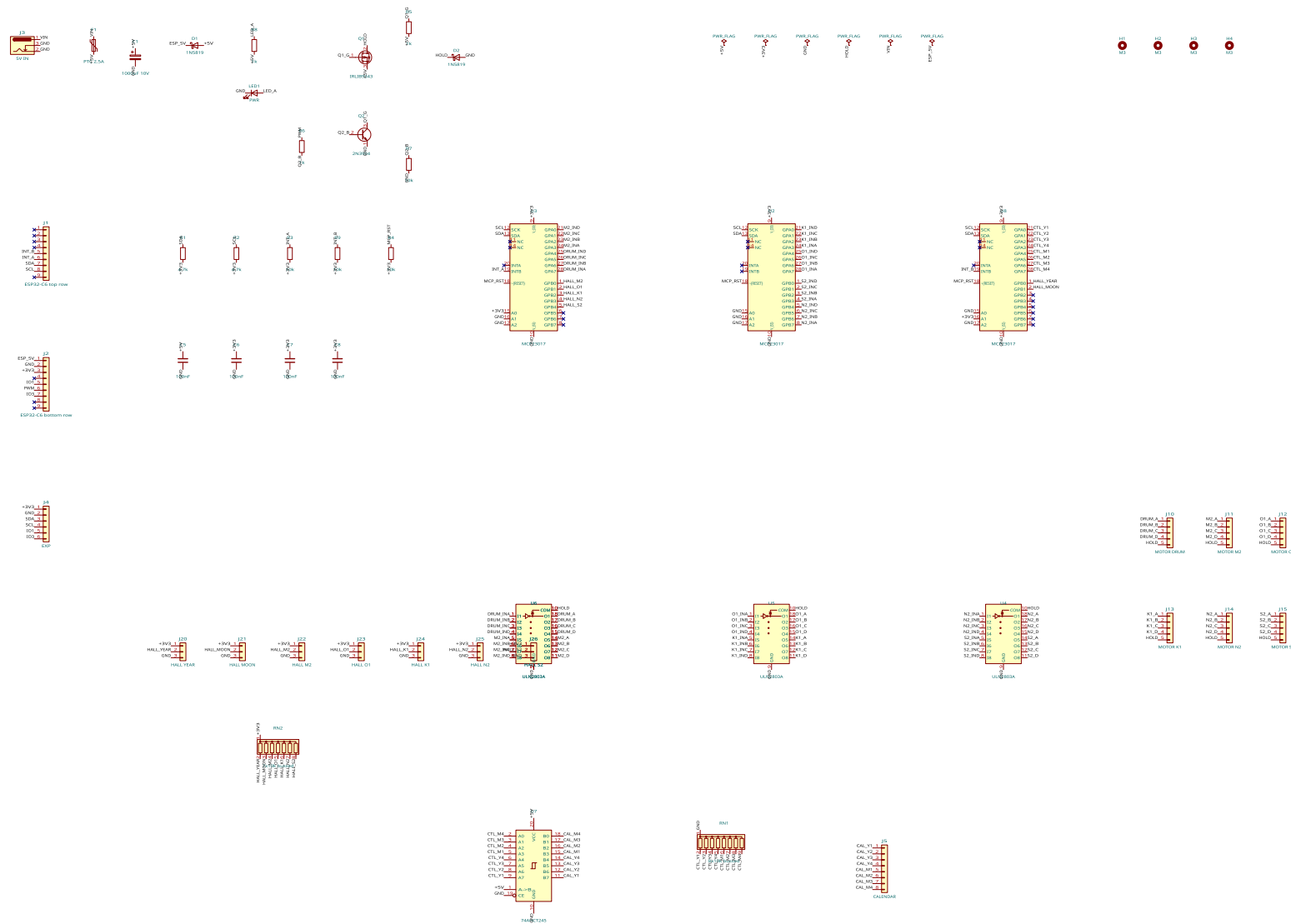
G2 · The main board, logically



Signal names, not GPIO numbers, are on the silkscreen. Hall inputs have 10 kΩ pull-ups (RN2); the calendar lines have pull-downs (RN1).

Every block and net from `pcb/design.py`, the same data the board is built from.

G3 · Main board, full schematic



The drawing is vector: zoom in as far as you like. Generated from `pcb/design.py` by `pcb/gen_schematic.py`; ERC clean.